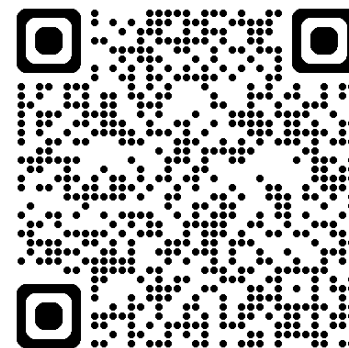




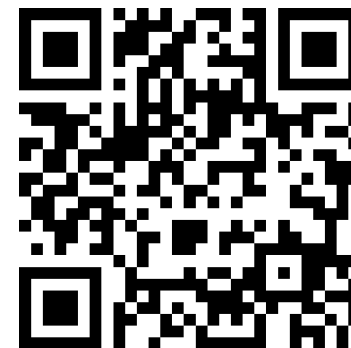
深度學習 Deep Learning

Transformers
自注意力機制模型

Instructor: 林英嘉 (Ying-Jia Lin)
2025/11/05



[Course GitHub](#)



[Slido # DL1105](#)

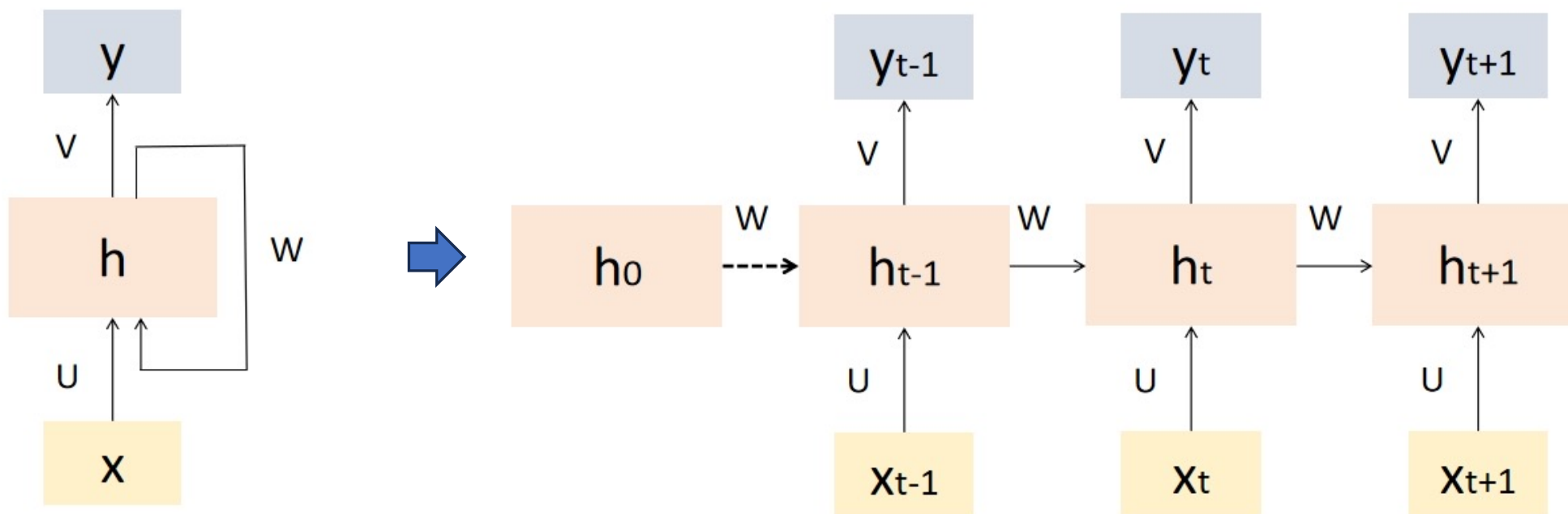
Outline

- Weakness of RNNs and CNNs [10 min]
- Transformers [90 min]
- Announcement about Term Project [10 min]
- Homework 3 [25 min]
- Quiz [15 min]

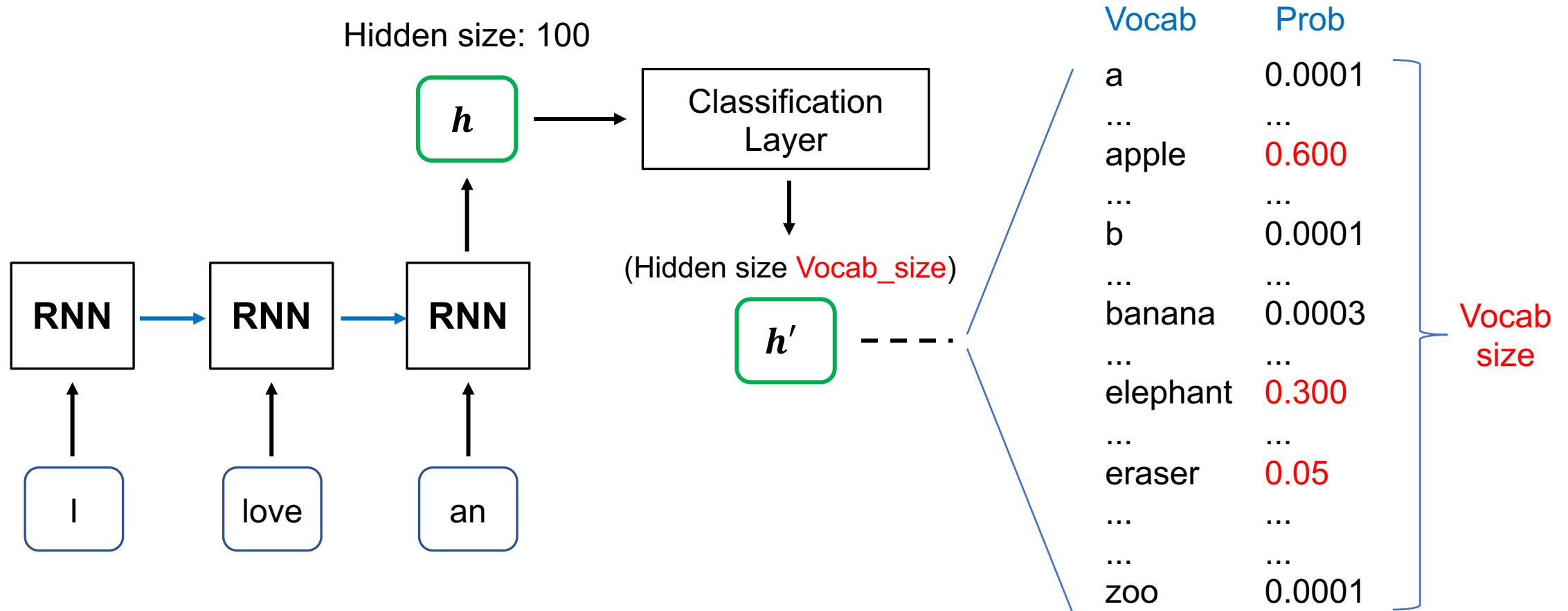


[Recap] Recurrent Neural Networks (RNN)

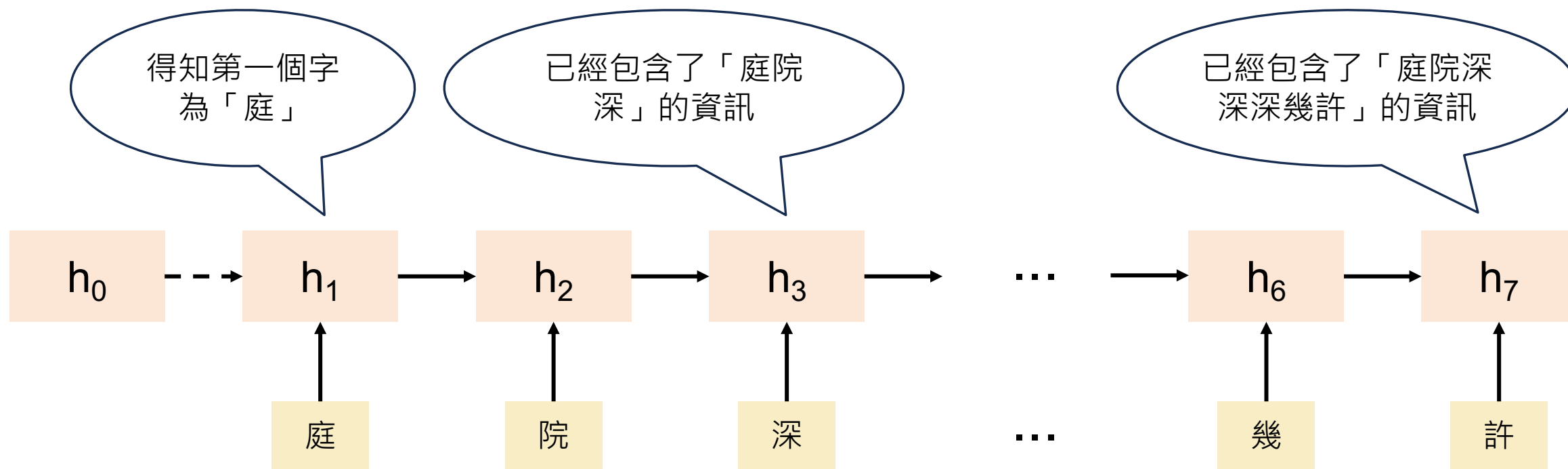
- Recurrent Neural Networks (RNNs) are a type of artificial neural network designed to handle sequential data by **capturing temporal dependencies**.
 - RNN 的“遞迴”是指它在每一個時間步中重複使用相同的參數（同一組權重），並把先前的隱藏狀態傳到下一個時間步



[Recap] Text Generation with RNN



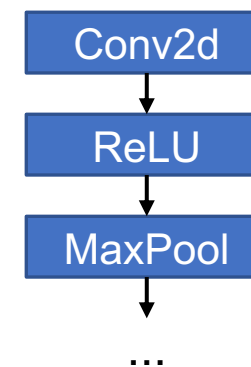
[Recap] RNN 的記憶力意義



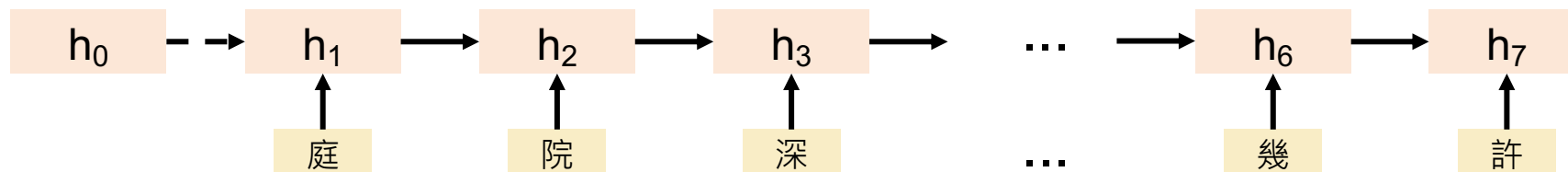
序列模型 (Sequential Model)

一層一層執行
(Recap)

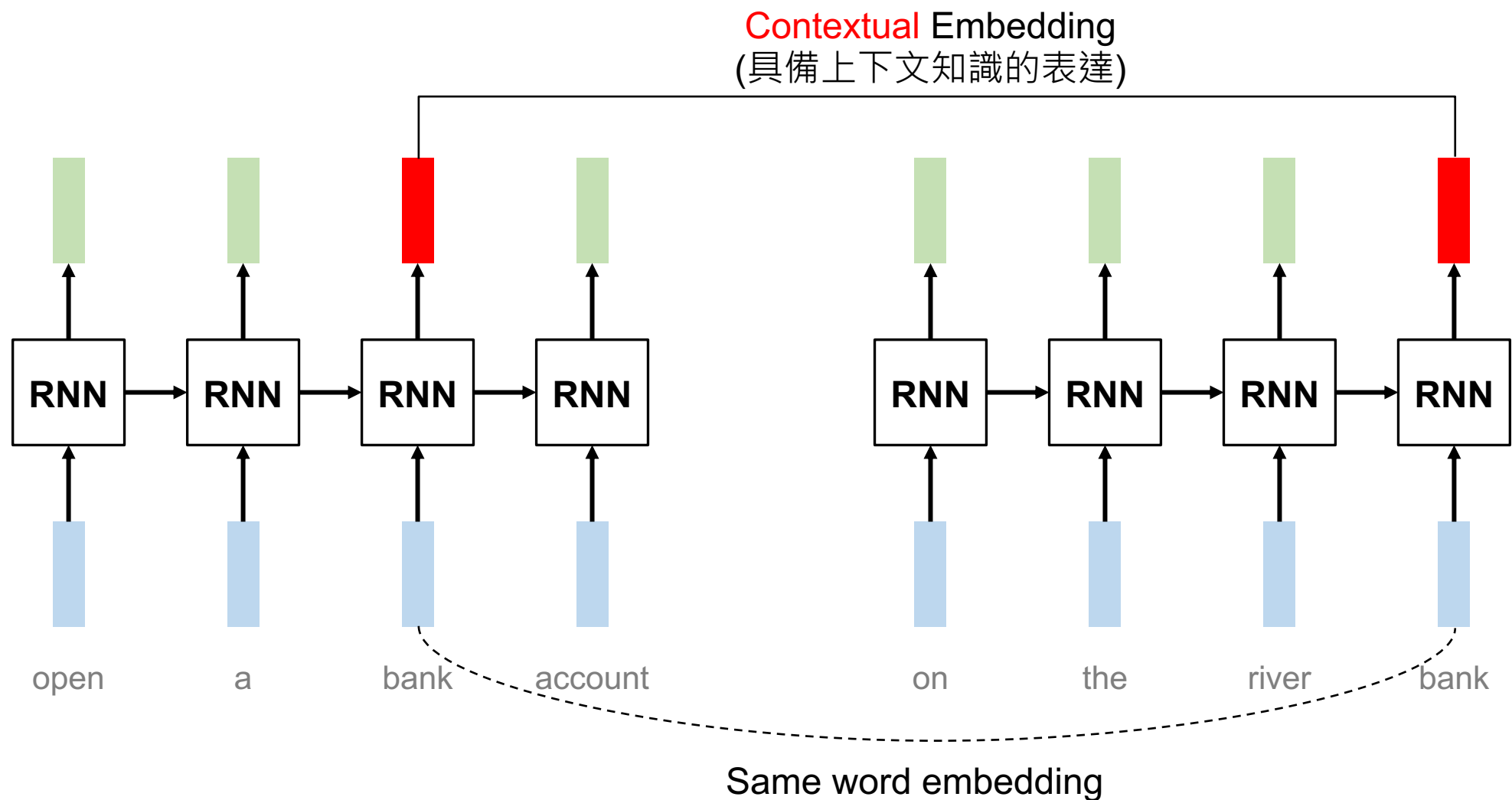
```
class SimpleCNN(torch.nn.Module):  
    """寫法2: 模型傳遞流程用 `nn.Sequential` 包起來  
    """  
    def __init__(self, num_classes=10, fc_hidden_size=256, dropout_ratio=0.5):  
        super().__init__()  
  
        self.feature_extractor = nn.Sequential(  
            nn.Conv2d(in_channels=3, out_channels=64, kernel_size=3, padding=1),  
            nn.ReLU(),  
            nn.MaxPool2d(kernel_size=2, stride=2),  
  
            nn.Conv2d(in_channels=64, out_channels=64, kernel_size=3, padding=1),  
            nn.ReLU(),  
            nn.MaxPool2d(kernel_size=2, stride=2),  
  
            nn.Conv2d(in_channels=64, out_channels=128, kernel_size=3, padding=1),  
            nn.ReLU(),  
            nn.MaxPool2d(kernel_size=2, stride=2),  
        )
```



序列模型：
一步一步執行，
有時間點關係

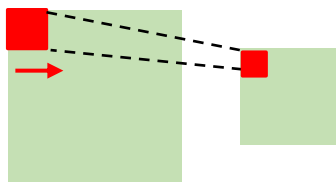


為什麼我們需要序列模型？

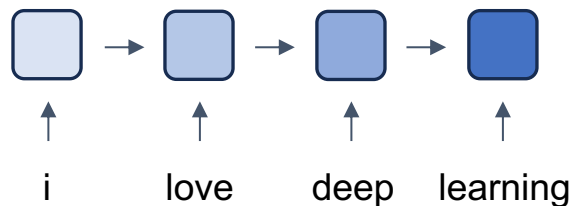


深度學習常見模型架構

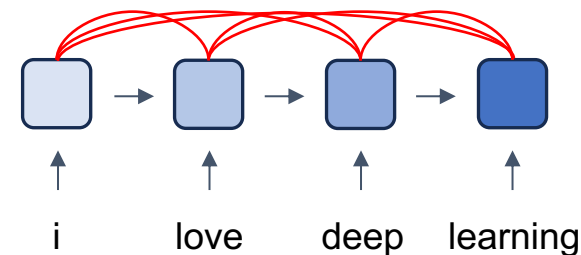
CNN (著重局部資訊)



RNN (著重記憶力)

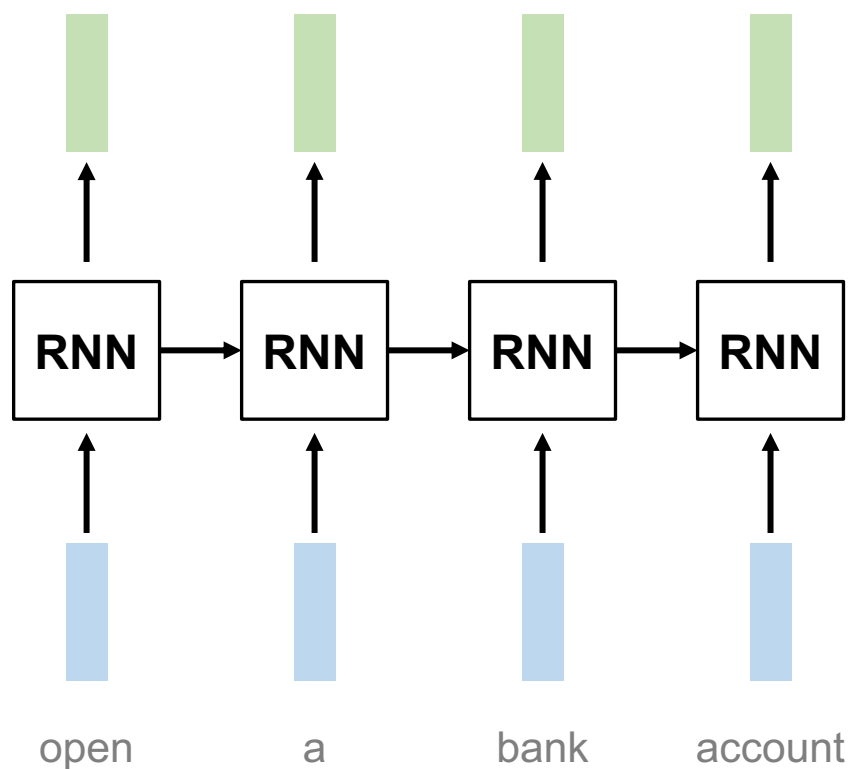


Transformer (著重全局資訊與記憶力)

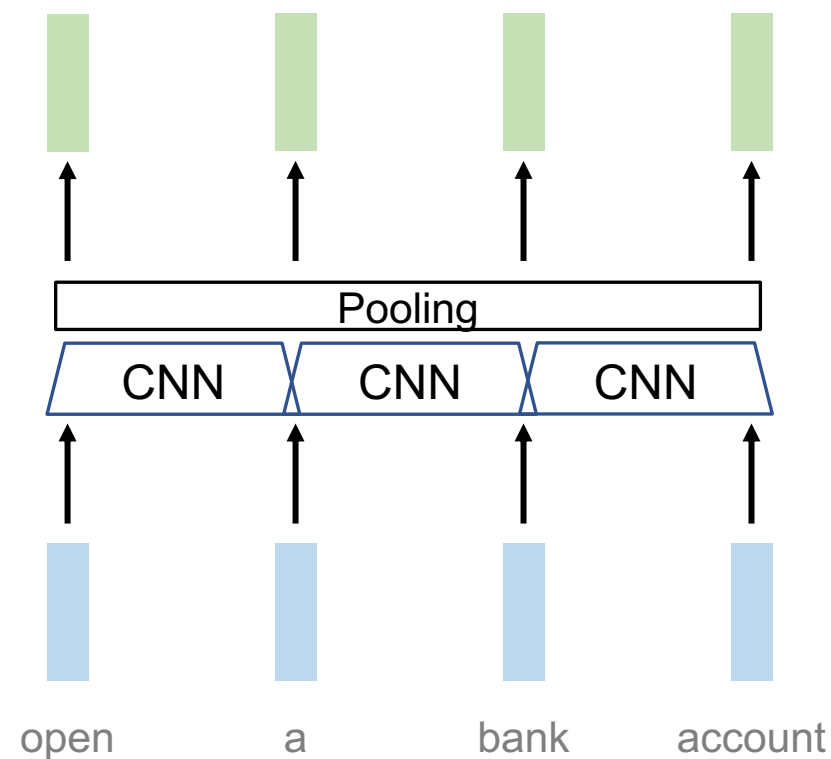


RNN 與 CNN 作為序列模型的問題

問題：RNN 難以平行化



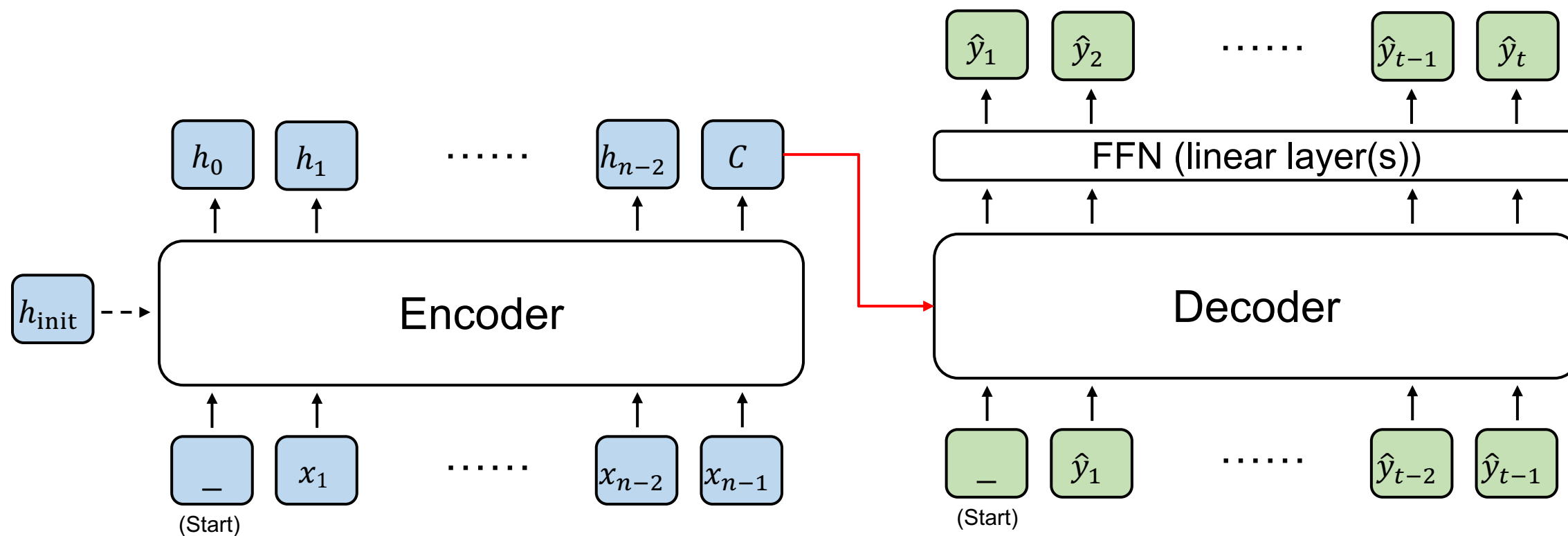
問題：CNN 著重局部資訊



Encoder and Decoder

輸出是 hidden states

輸出是 sequence (文字)

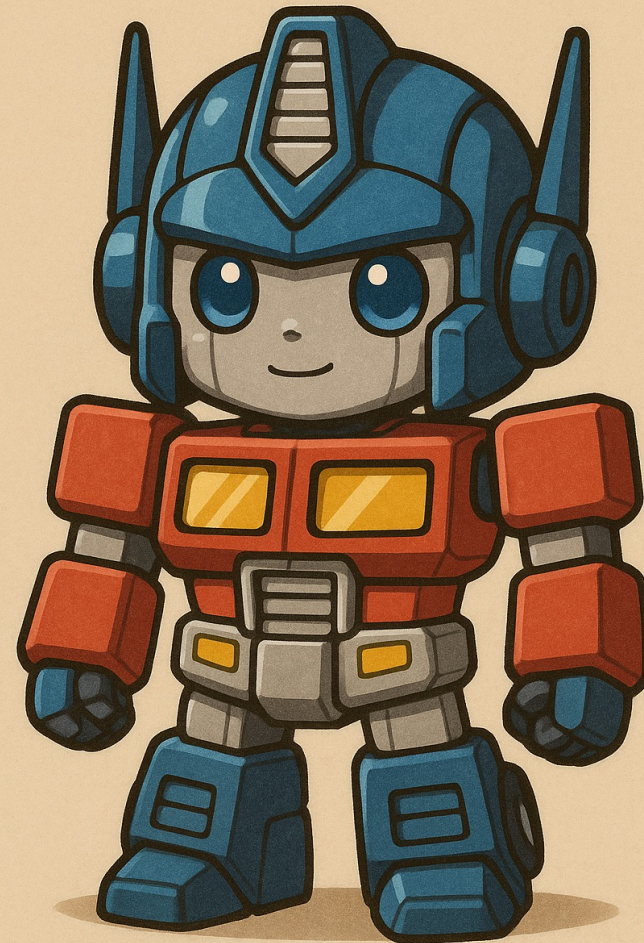


Input sequence

(Encoder 的輸出有很多種
被Decoder運用的方式)



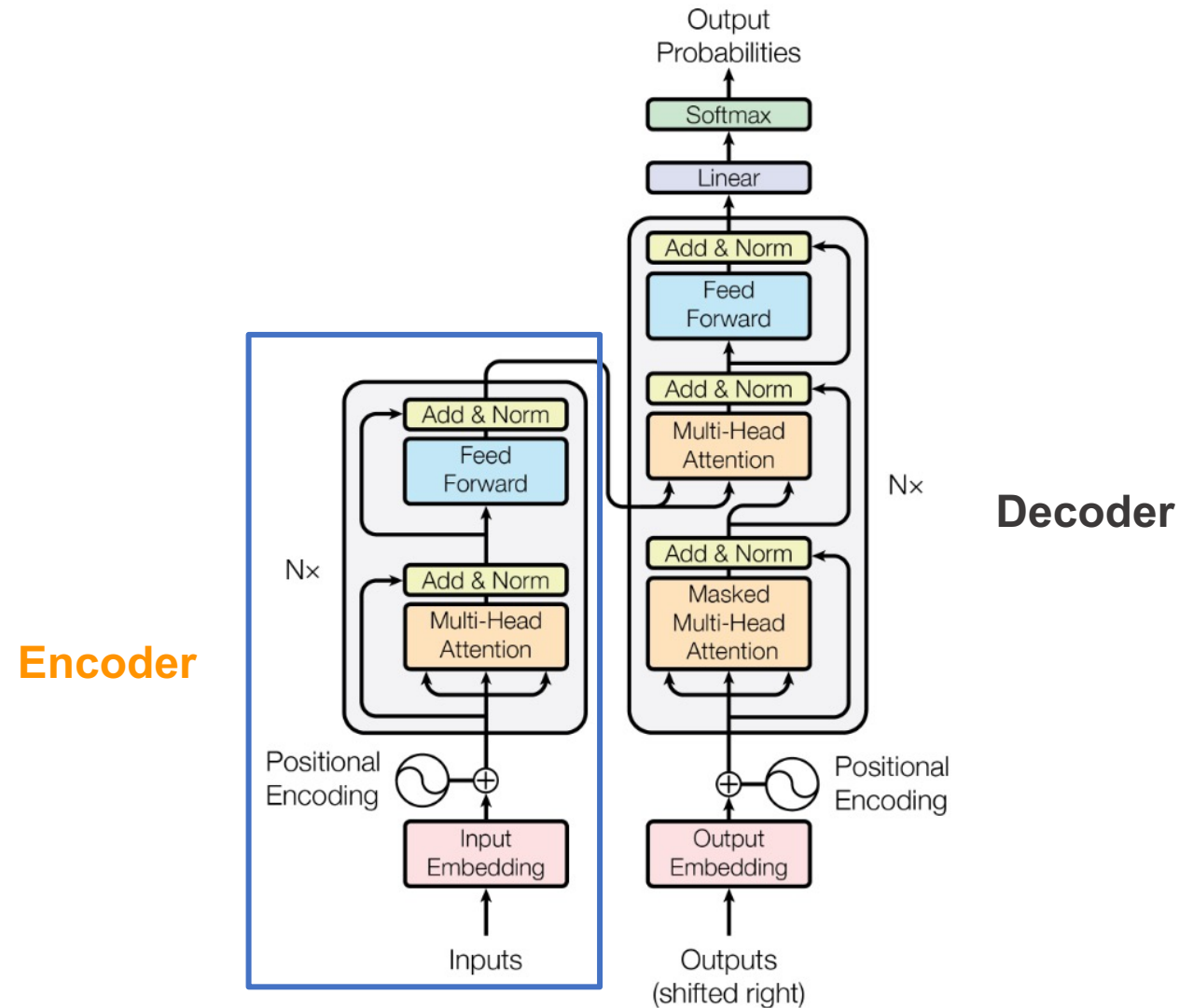
Transformer



TRANSFORMER

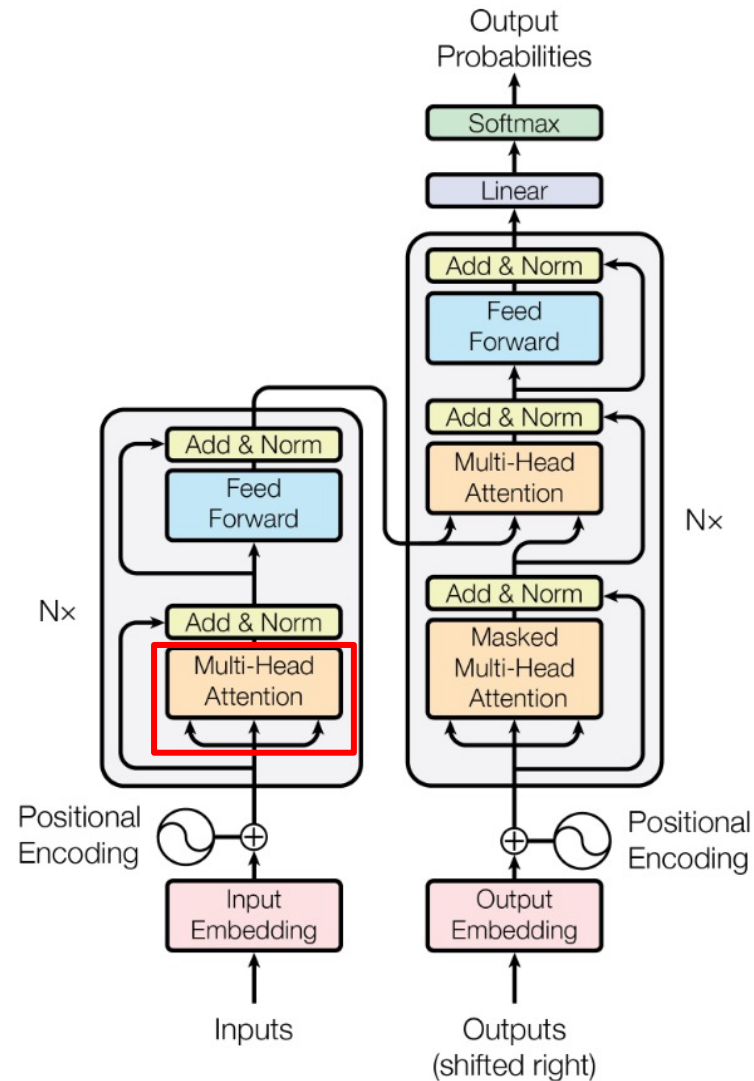
Overview of Transformer

Vaswani, Ashish, et al. "Attention is all you need." NeurIPS (2017).



Outline of Transformer

Vaswani, Ashish, et al. "Attention is all you need." NeurIPS (2017).

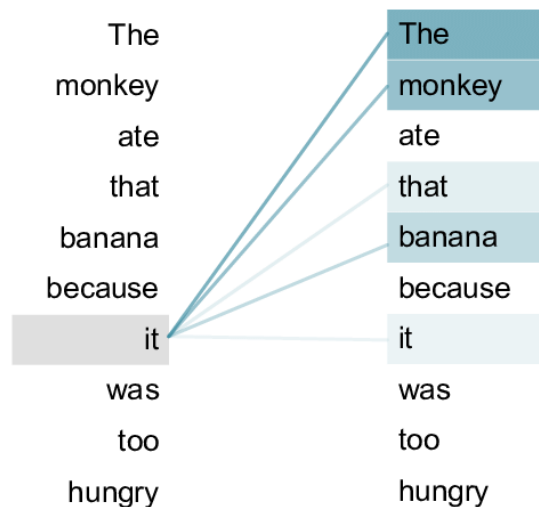


Transformer

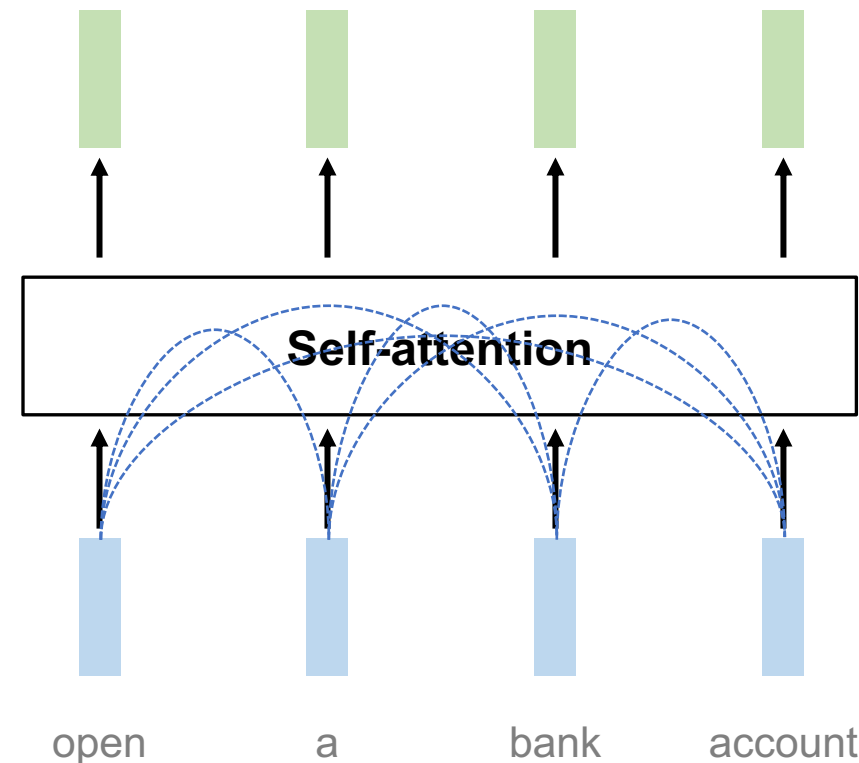
Source: [An example of the self-attention mechanism following long-distance... | Download Scientific Diagram \(researchgate.net\)](#)

- Transformer 來自論文：Attention Is All You Need (Vaswani et al., 2017)
- Transformer 內部的主要機制為 Self-attention

Self-attention: 句子內的每個 token 自己
跟自己做 attention

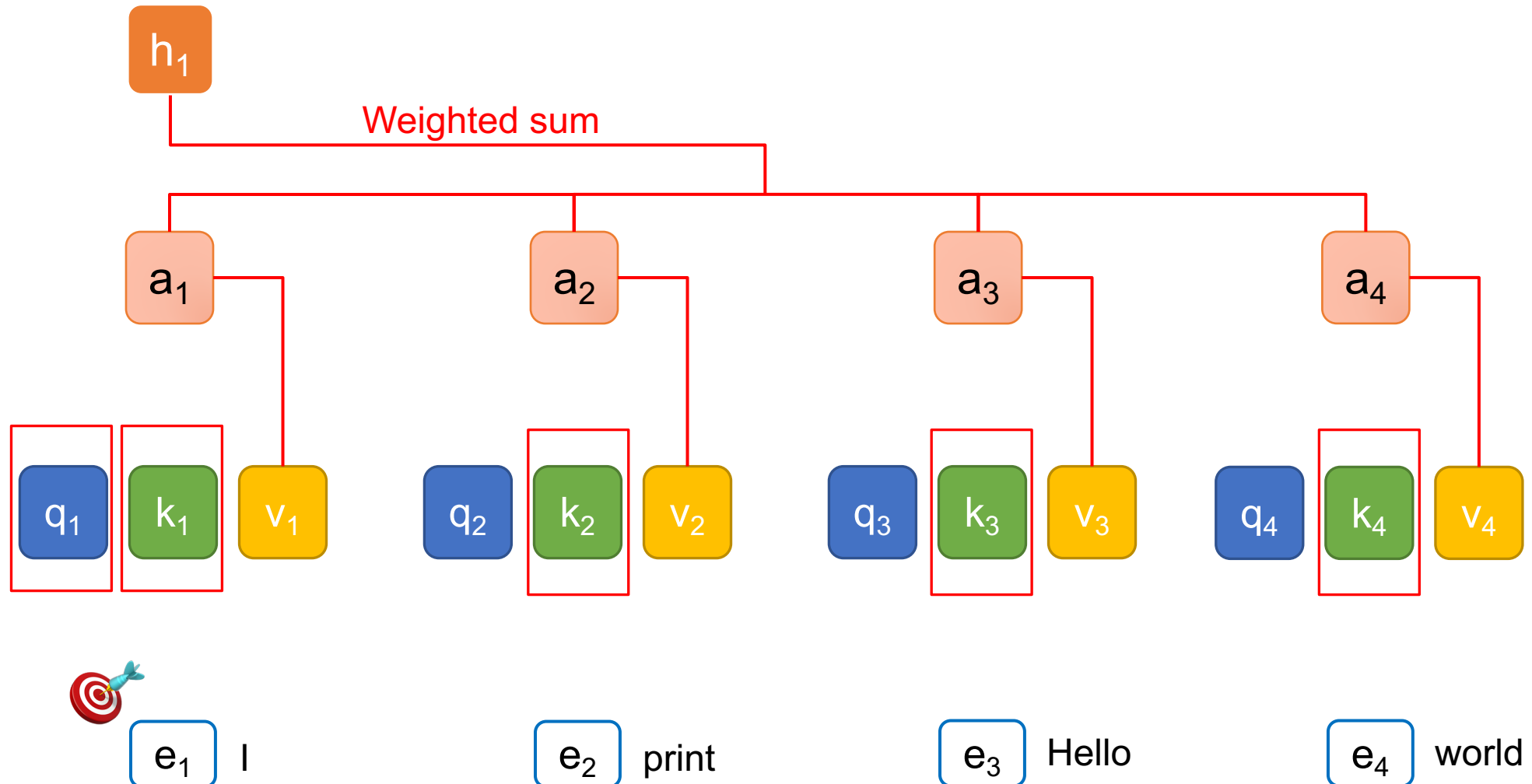


attention 進行的過程需要主動與被動



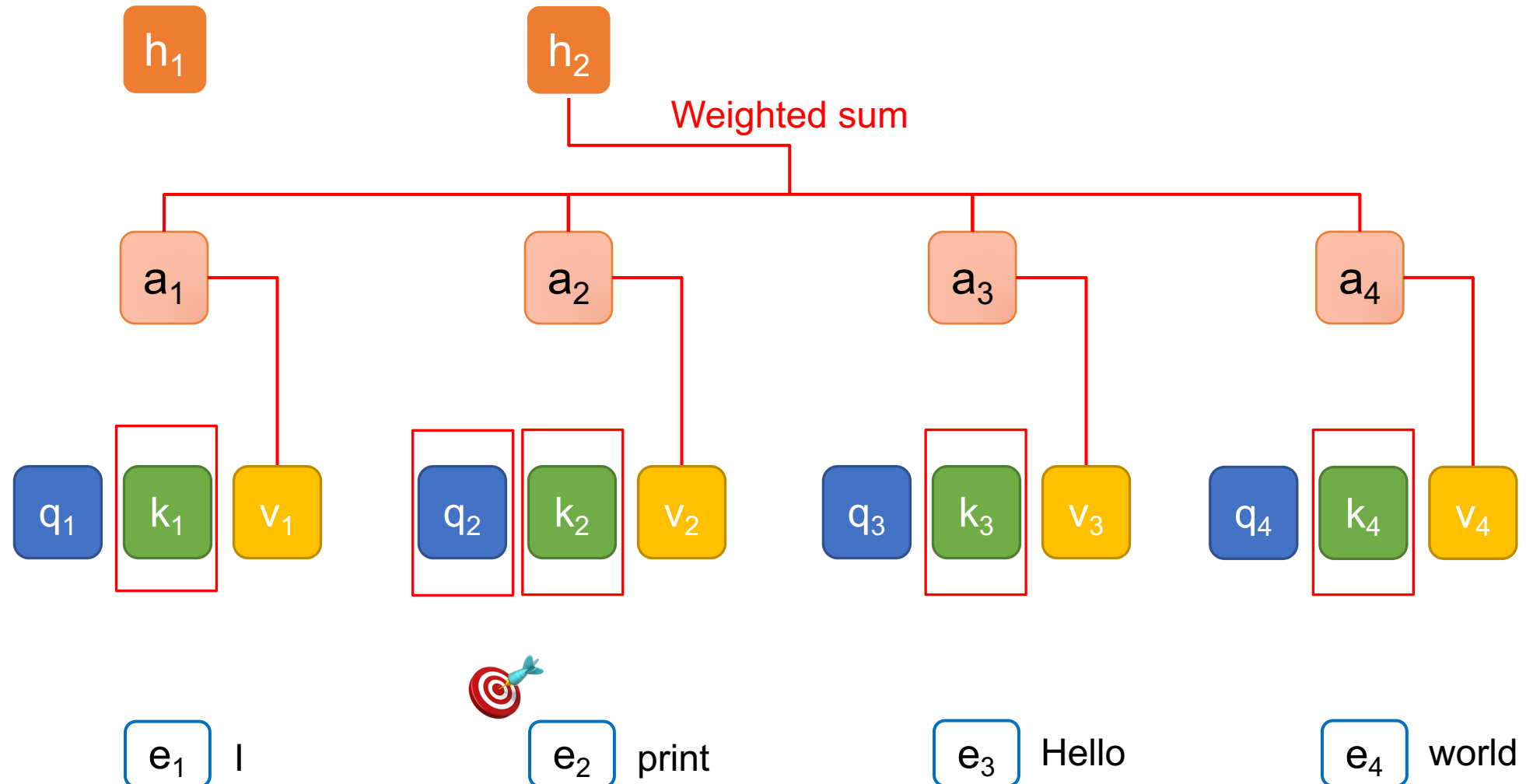
Query (Q), Key (K), and Value (V)

$t = 1$



Query (Q), Key (K), and Value (V)

$t = 2$



QKV 的比喻

Vaswani, Ashish, et al. "Attention is all you need." NeurIPS (2017).

Query

The image shows a YouTube search interface. At the top, the search bar contains the text "how to become a data scientist", which is highlighted with a red box and labeled "Query". Below the search bar, there are navigation tabs: "全部", "Shorts", "影片", "未觀看", "已觀看", "最新上傳", "直播", and "訂閱". The main content area displays a video titled "The Complete Data Science Roadmap" by "Programming with Mosh". The video thumbnail shows a man with glasses and a beard, with the text "DATA SCIENCE COMPLETE ROADMAP" and a duration of "6:13". The video title "The Complete Data Science Roadmap" is highlighted with a green box and labeled "Key". Below the title, the video description is highlighted with a green box and labeled "Value". The description reads: "Go from zero to a data scientist in 12 months. This step-by-step roadmap covers the essential skills you mu...". Below the description, there is a list of 11 chapters: "Introduction | Programming Languages | Version Control | Data Structures ...".

Value

- Query: (主動) 查詢的關鍵字；Key: (被動) 查詢的對象
- Value: 值，關鍵字與對象的匹配程度

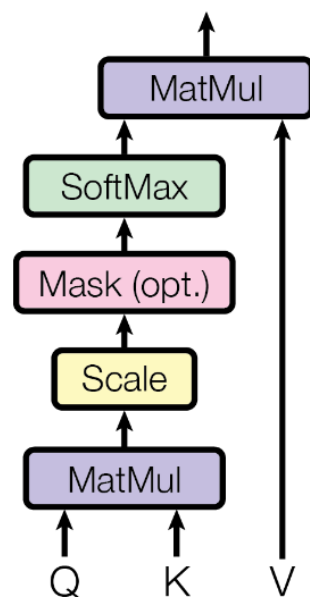


為什麼 Transformers 可以平行化？

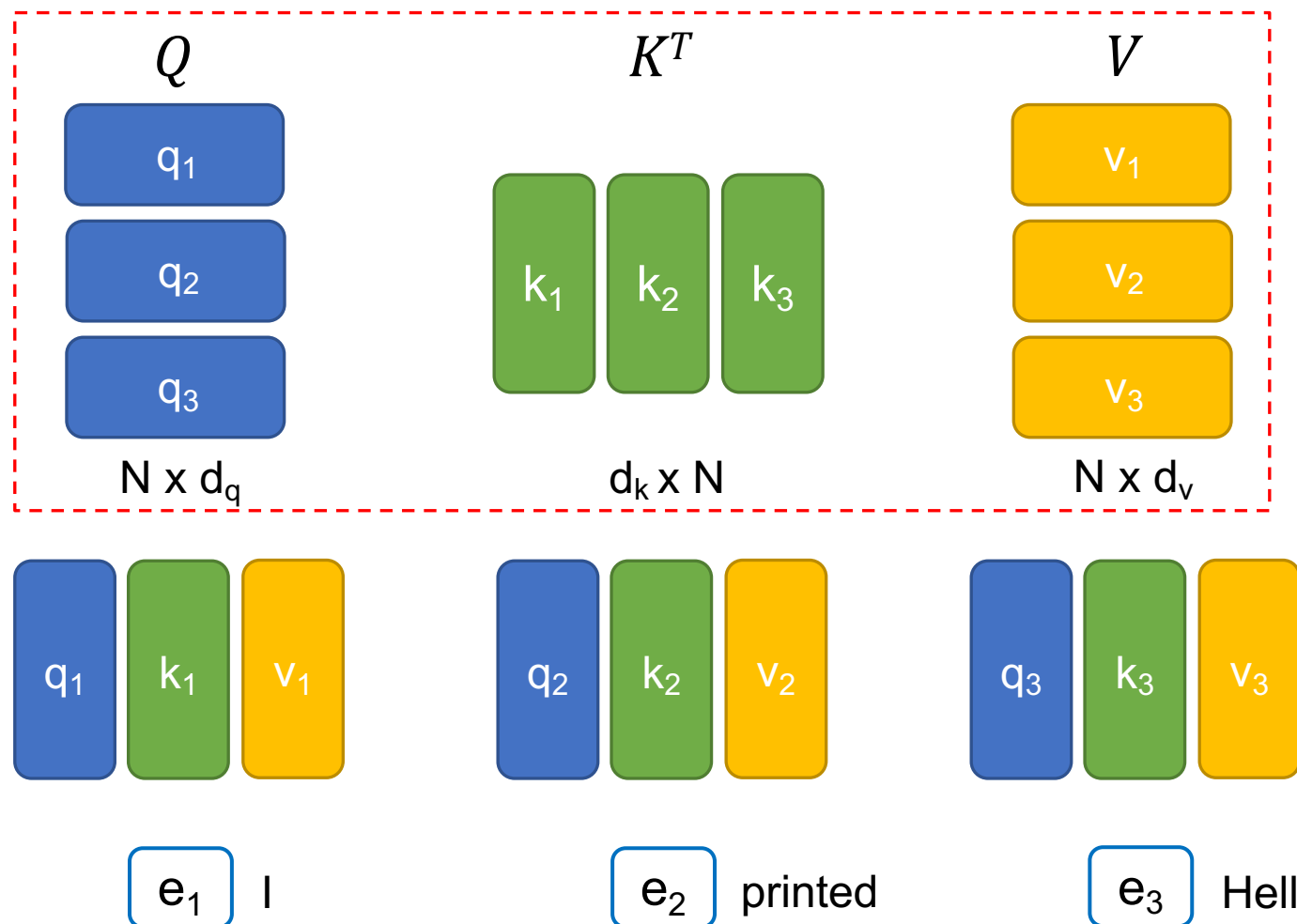
$$d_q = d_k = d_v$$

(自注意力機制可以利用矩陣乘積來進行平行化計算)

Scaled Dot-Product Attention

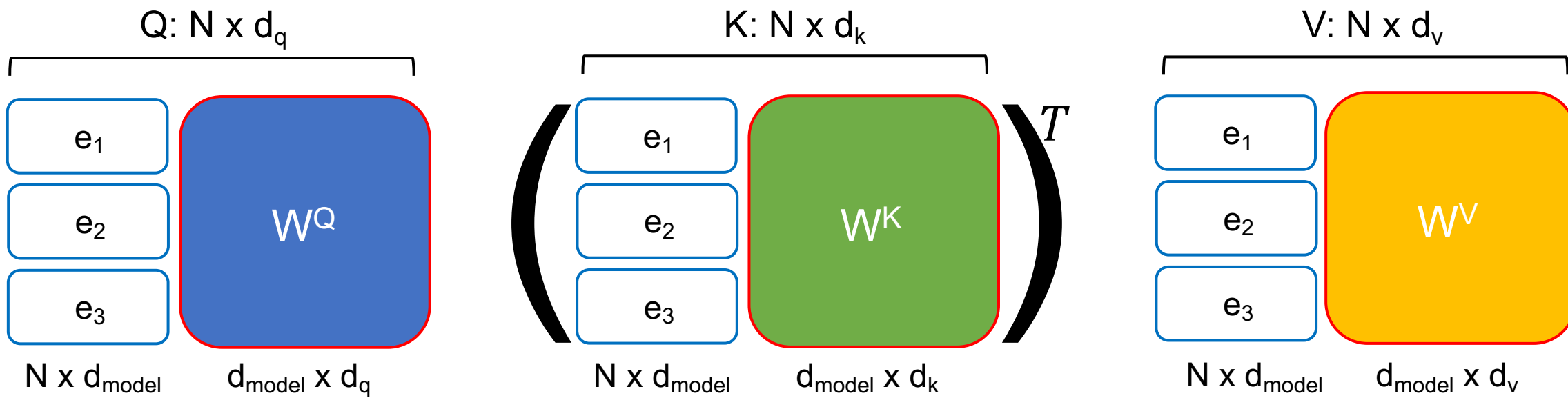


$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

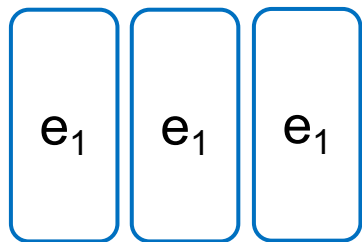


Transformers 的權重值 (1/2)

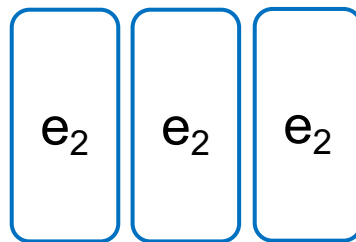
$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$



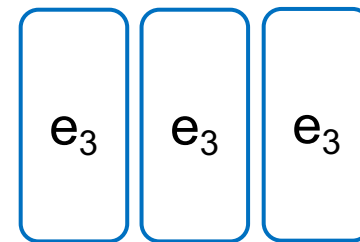
d_{model} :
Embedding
的維度大小



I



printed

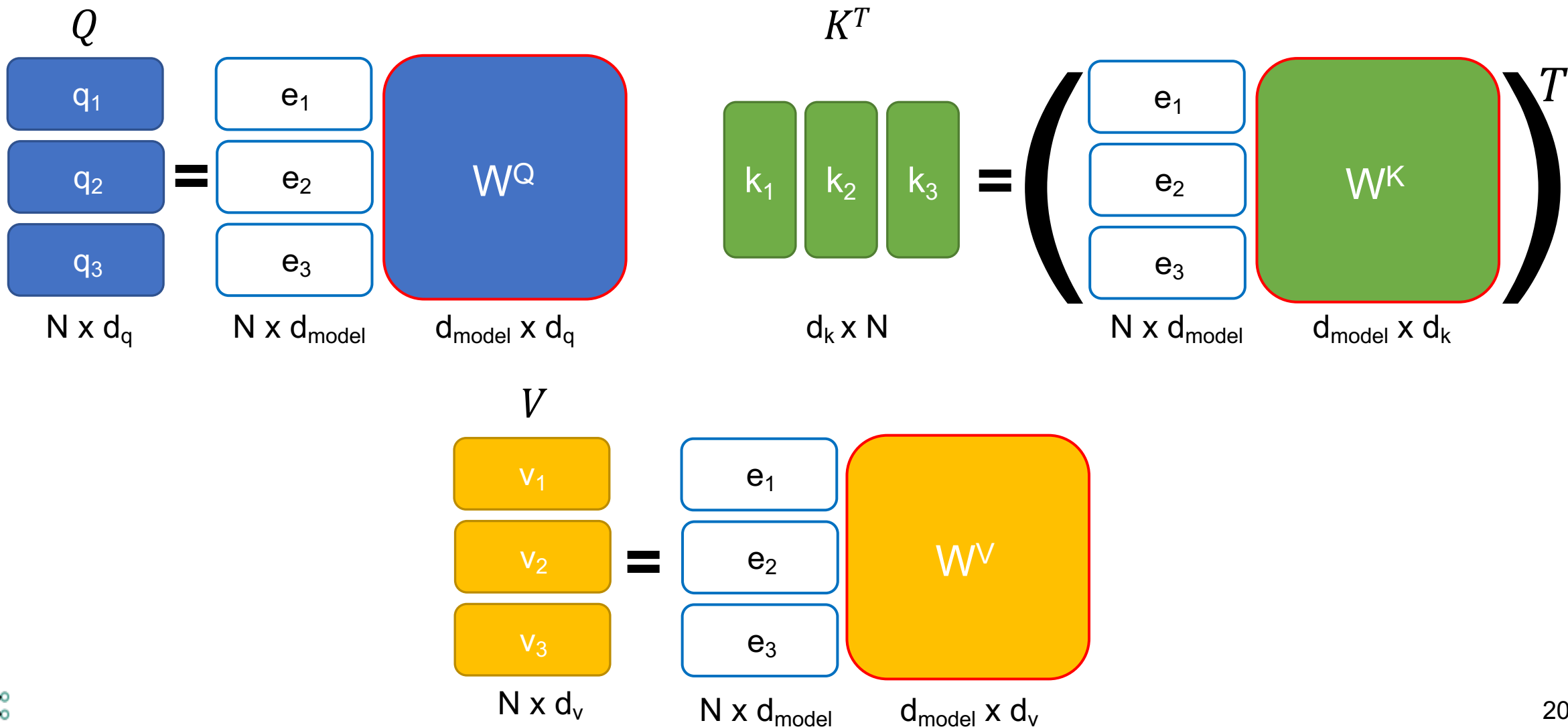


Hello



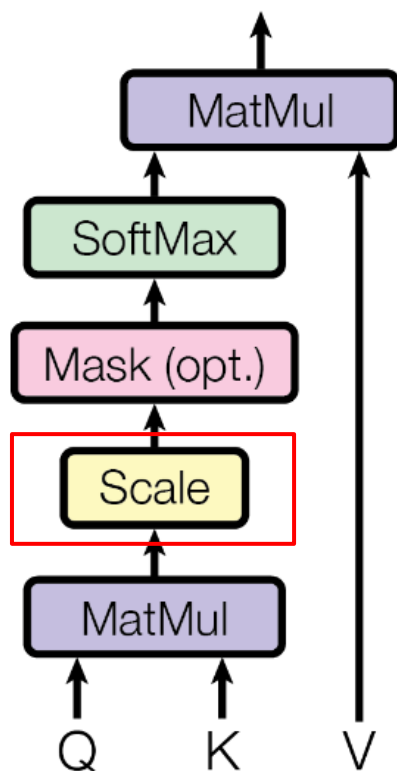
Transformers 的權重值 (2/2)

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$



Self-attention 的公式與縮放項

Scaled Dot-Product Attention



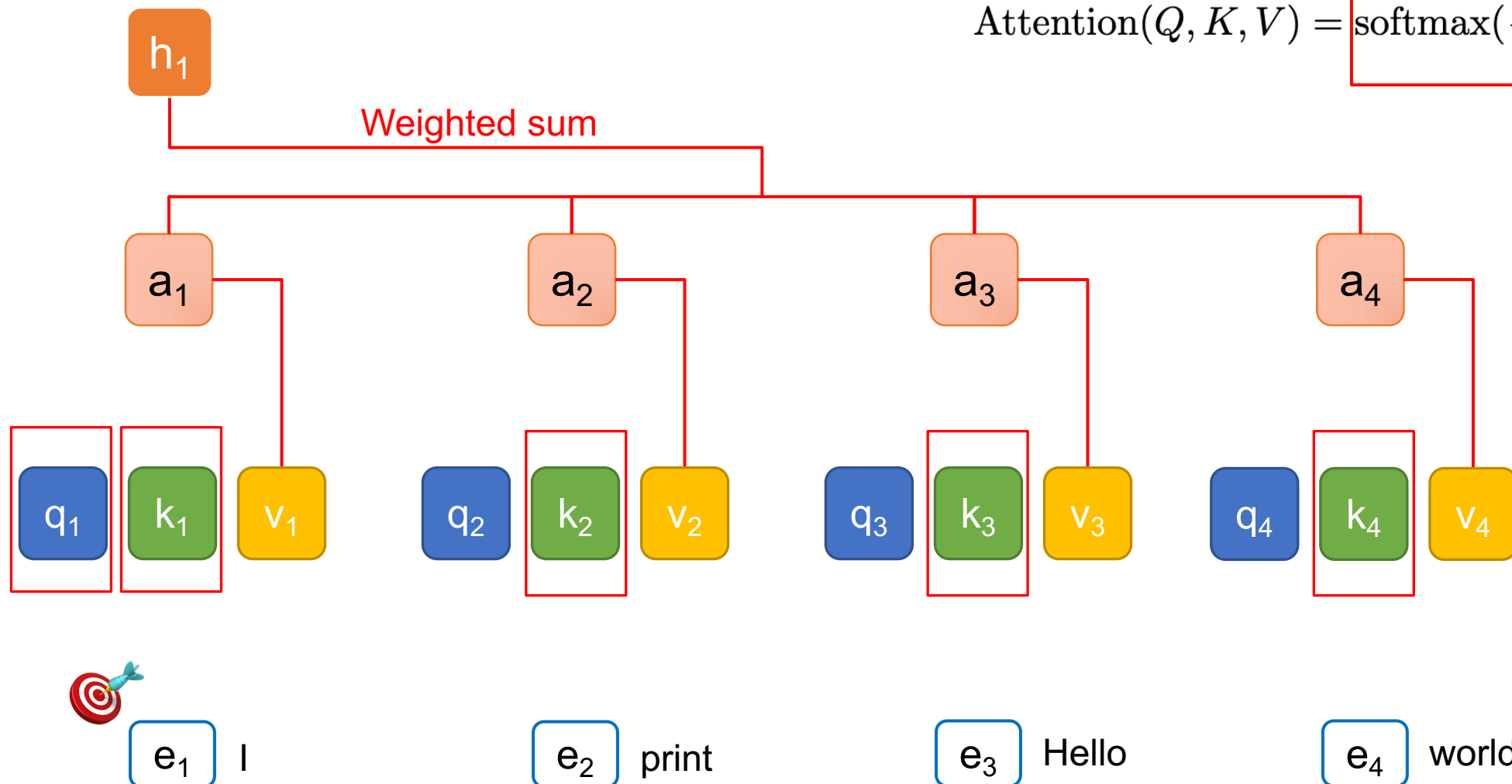
$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$



Query (Q), Key (K), and Value (V)

t = 1

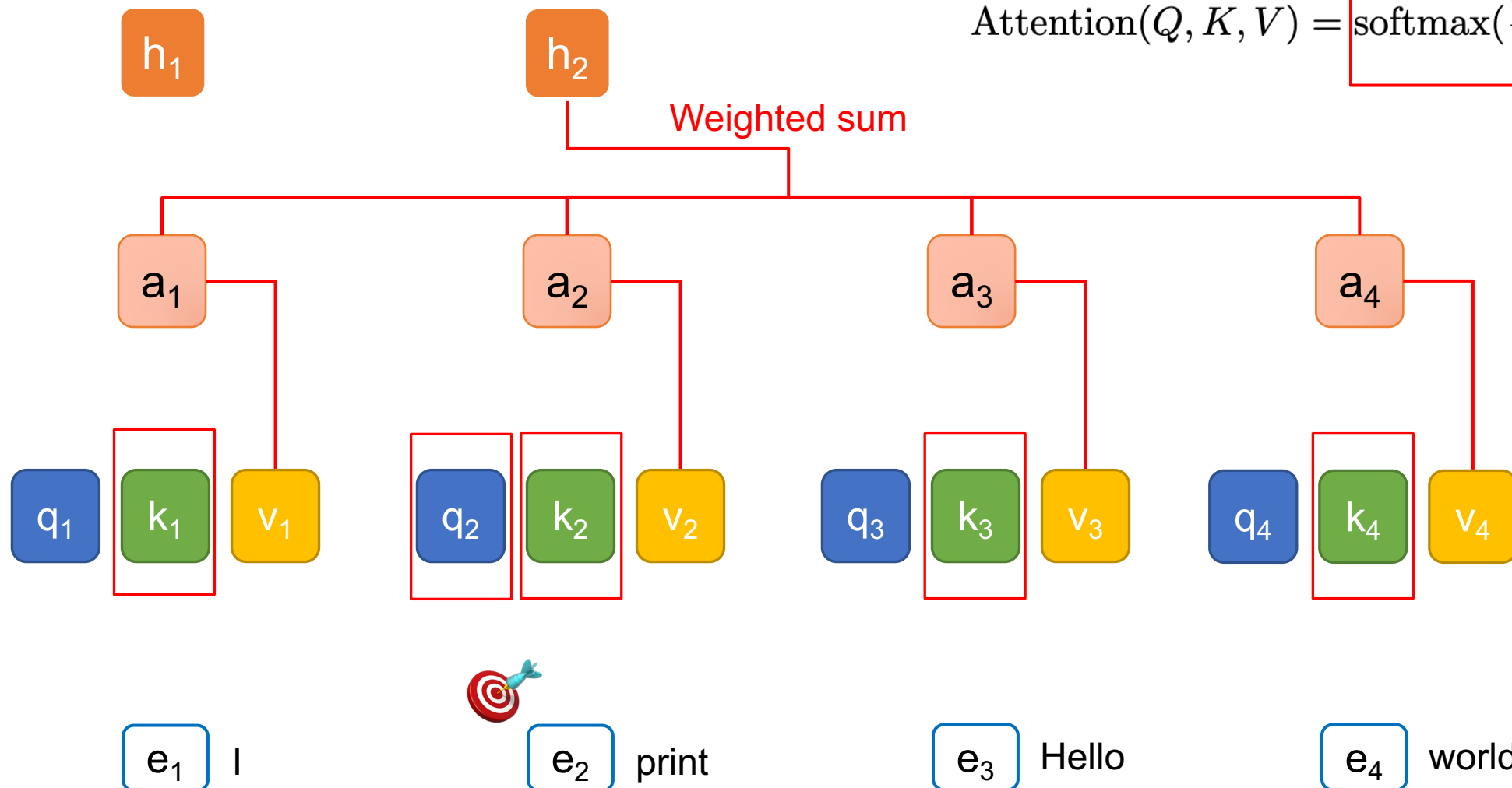
$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$



Query (Q), Key (K), and Value (V)

t = 2

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$



Self-attention 動畫

Figure source:
<https://research.google/blog/transformer-a-novel-neural-network-architecture-for-language-understanding/>



Attention Is All You Need

Essential

Ashish Vaswani*
Google Brain
~~avaswani@google.com~~

Noam Shazeer*

Google Brain
~~noam@google.com~~

Anthropic

Niki Parmar*

Google Research
~~nikip@google.com~~

Inceptive

Jakob Uszkoreit*

Google Research
~~usz@google.com~~

Cohere

Aidan N. Gomez* †

University of Toronto
~~aidan@cs.toronto.edu~~

Łukasz Kaiser*

Google Brain
~~lukaszkaizer@google.com~~

OpenAI

Sakana AI

Llion Jones*

Google Research
~~llion@google.com~~

NEAR

Illia Polosukhin* ‡

~~illia.polosukhin@gmail.com~~

<https://arxiv.org/abs/1706.03762v6>

Multi-Head Attention (MHA)

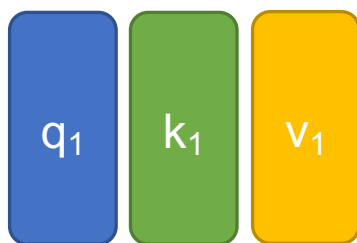
- 到目前為止，我們已經介紹了 self-attention
 - We **query** the search engine
 - We search engine tries to map our query to the **keys**
 - The outputs are the attended **values** (e.g., the best matched videos).
- What if we want to have queries focusing on different aspects?
 - For example, one set of queries focusing on semantic similarity, another set focusing on passive/active voice.
 - **We need multiple attention mechanisms, i.e. multiple (attention) heads!**



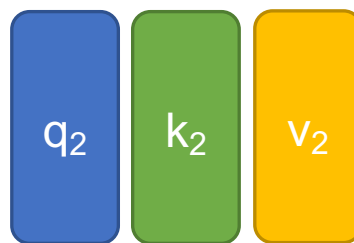
Multi-Head Attention (MHA)

上標: head
下標: token index

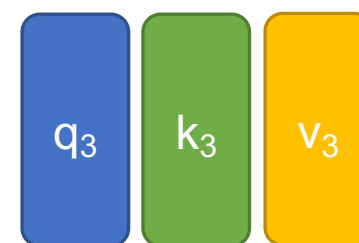
Head = 1 (without MHA)



e_1 |

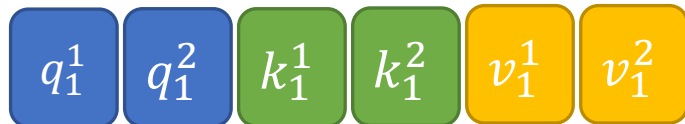


e_2 printed

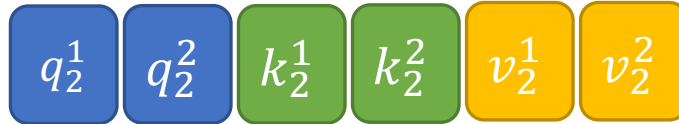


e_3 Hello

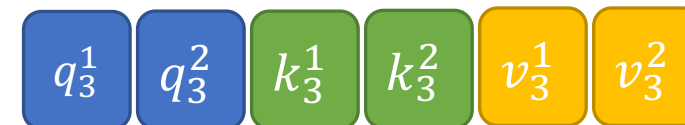
Head = 2 (with MHA)



e_1 |



e_2 printed

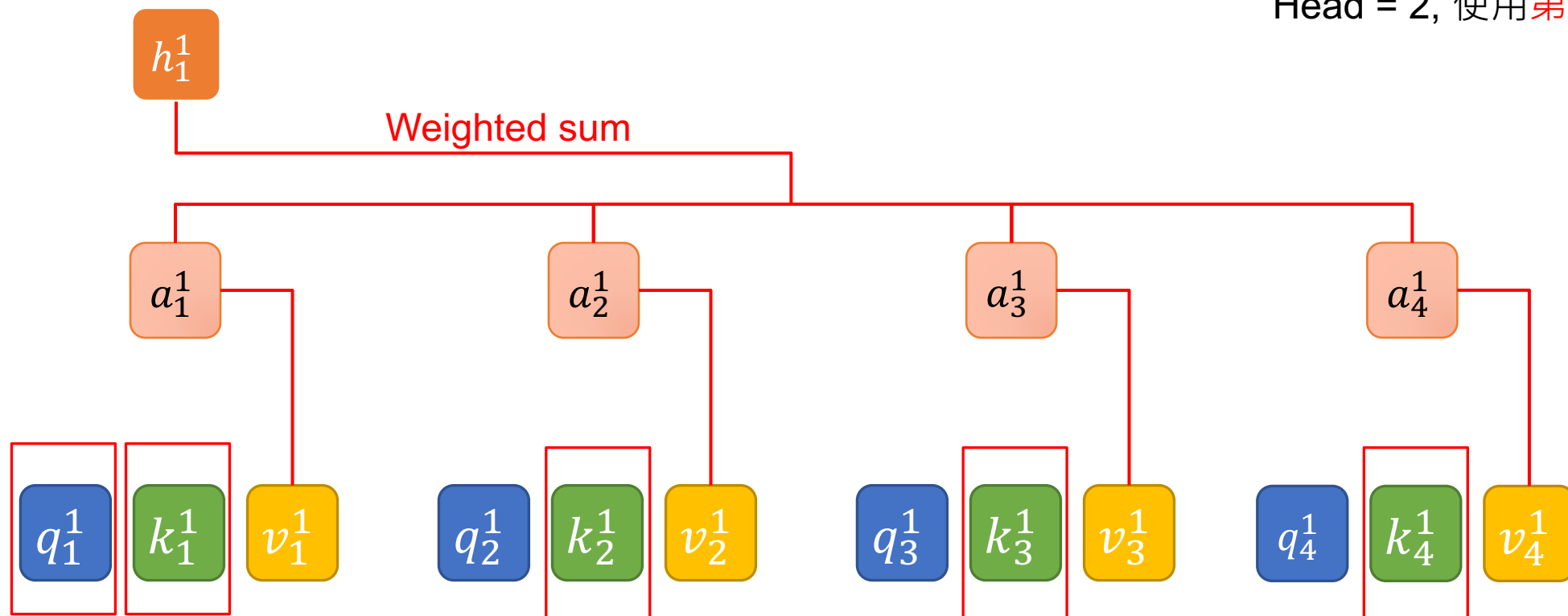


e_3 Hello



Query (Q), Key (K), and Value (V) with MHA

Head = 2, 使用第一個 head



e_1 I

e_2 printed

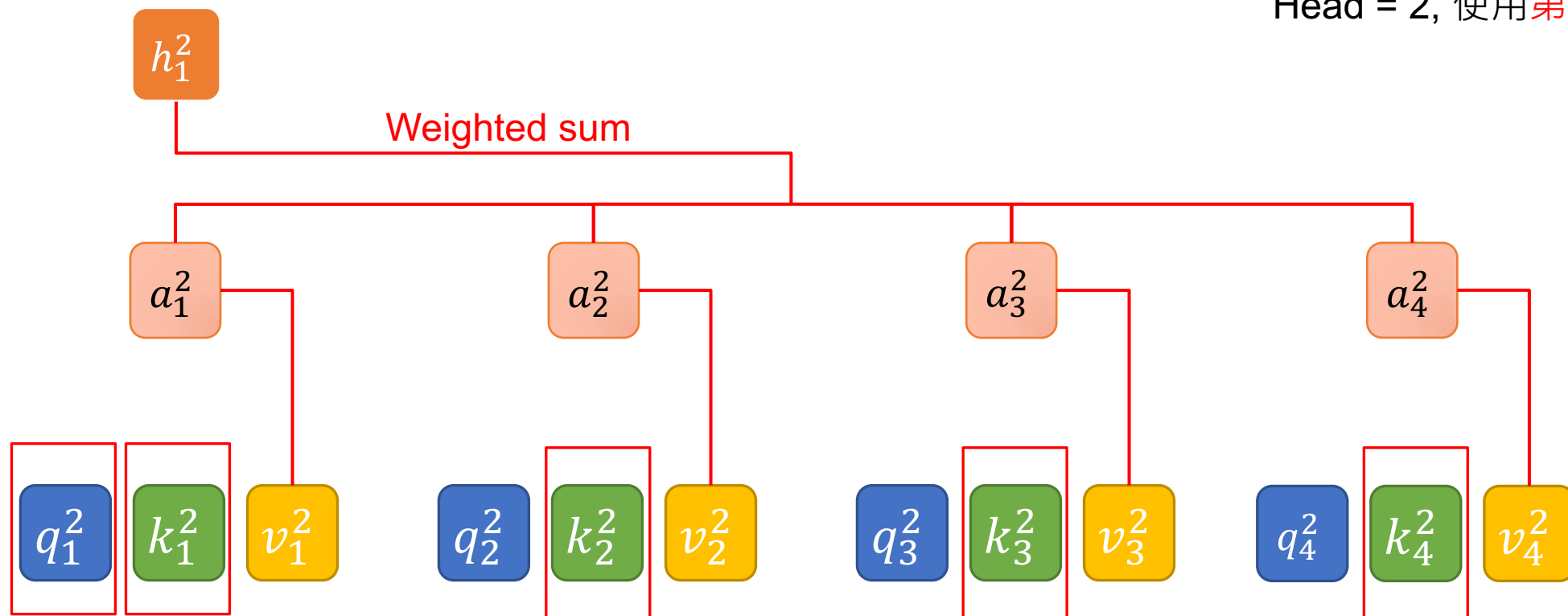
e_3 Hello

e_4 world



Query (Q), Key (K), and Value (V) with MHA

Head = 2, 使用第二個 head



e_1 I

e_2 printed

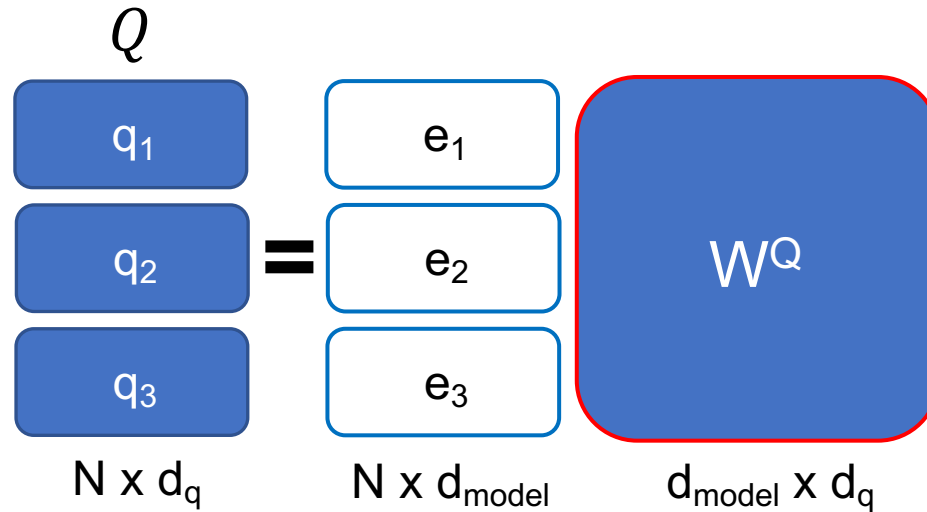
e_3 Hello

e_4 world

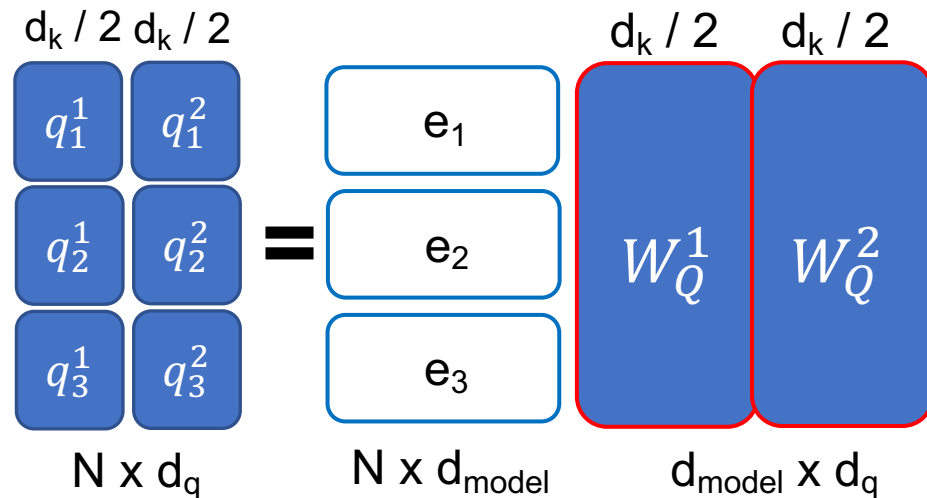


Input Dimensions of MHA

Head = 1 (without MHA)



Head = 2 (with MHA)



Output Dimensions

$$\text{Multihead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O$$

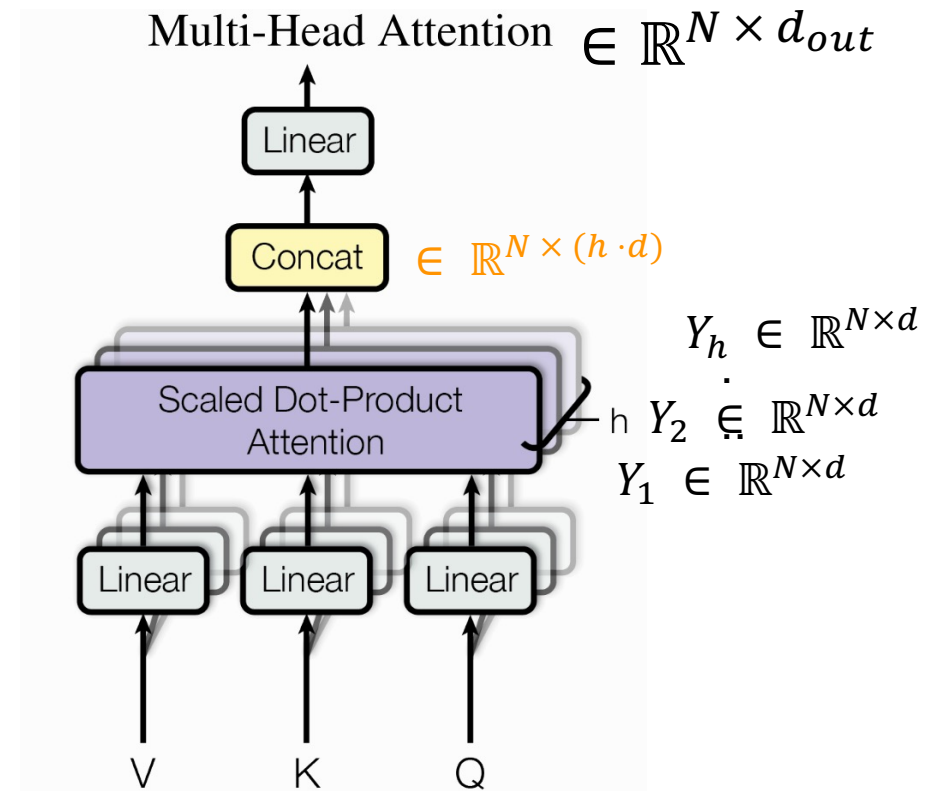
$\in \mathbb{R}^{N \times (h \cdot d)}$
 $\in \mathbb{R}^{(h \cdot d) \times d_{out}}$

Head = 1 (without MHA)

$$h_1 \times W^O = h_1$$

Head = 2 (with MHA)

$$\begin{bmatrix} h_1^1 \\ h_1^2 \end{bmatrix} \times W^O = h_1$$



MHA 比較好嗎？

Clark, Kevin, et al. "What Does BERT Look at? An Analysis of BERT's Attention." Proceedings of the 2019 ACL Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP. 2019.

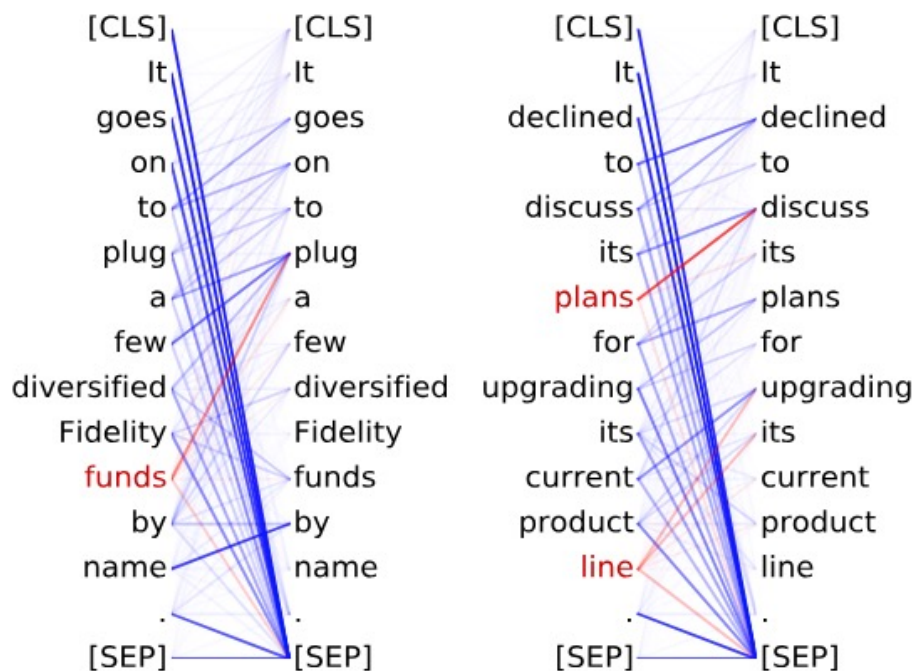
實驗證明：不同 head 關注的地方不太一樣

<layer>-<head number>

Head 8-10

關注受詞

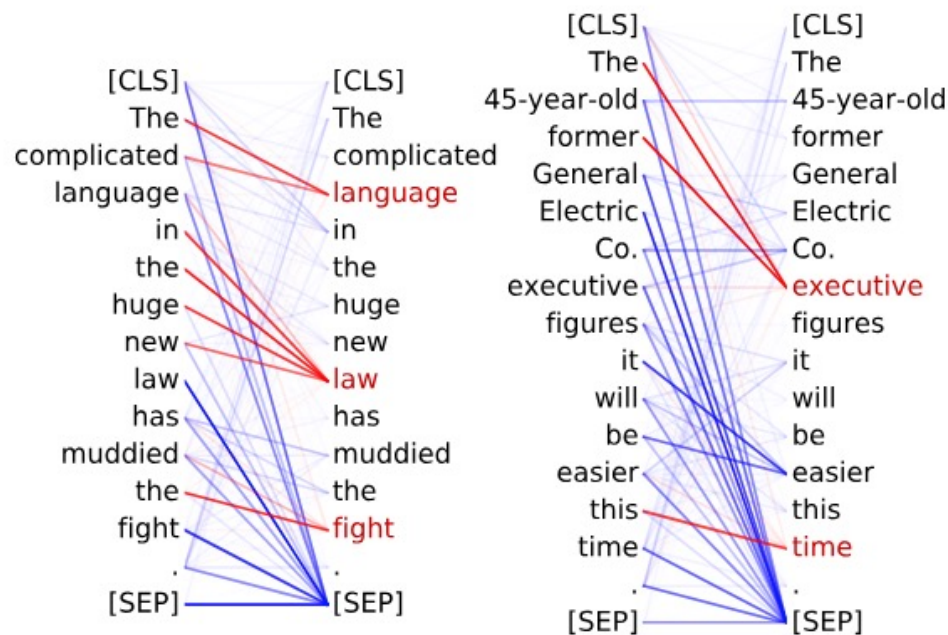
- **Direct objects** attend to their verbs
- 86.8% accuracy at the **dobj** relation



Head 8-11

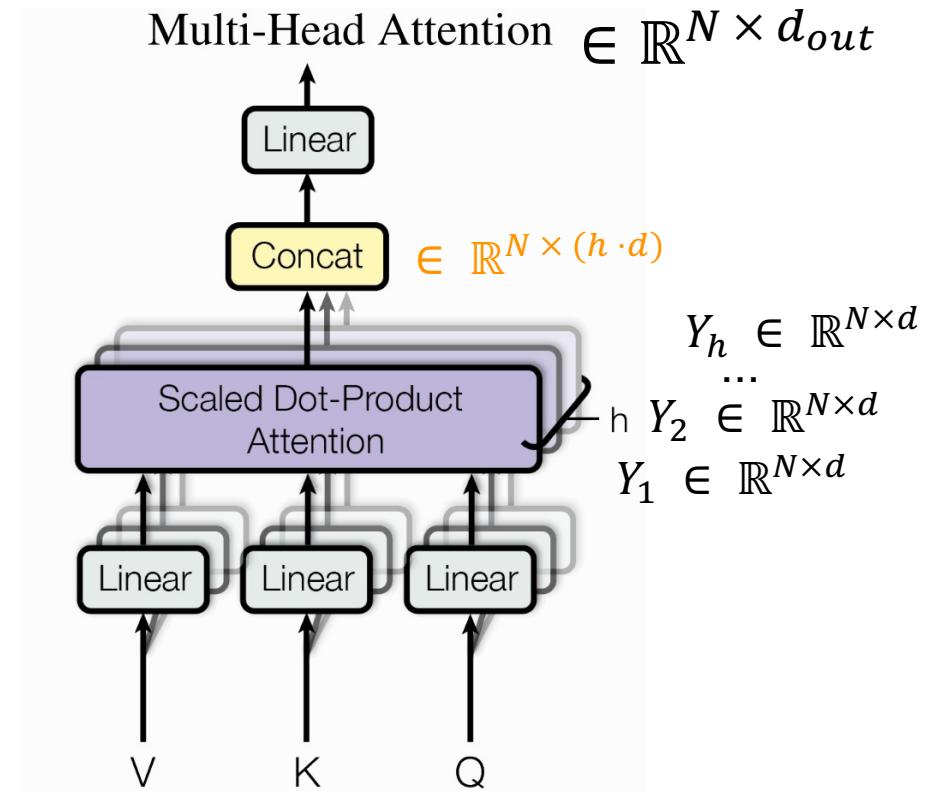
關注限定詞

- **Noun modifiers** (e.g., determiners) attend to their noun
- 94.3% accuracy at the **det** relation



Summary of MHA

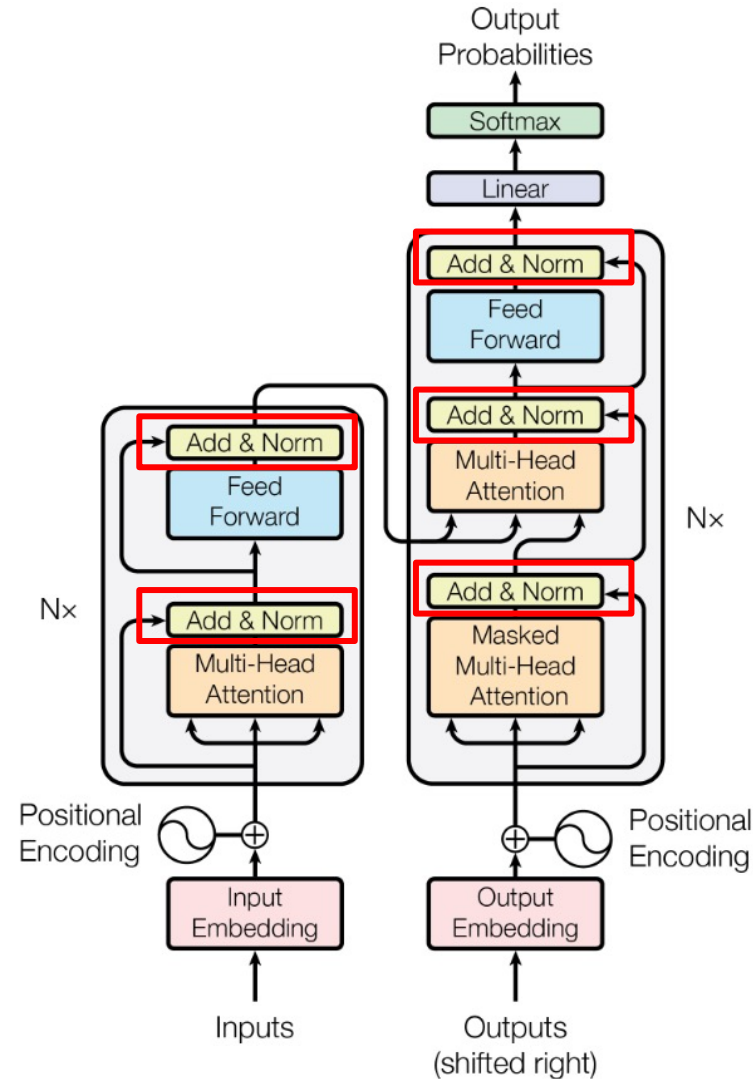
- With W^O , the concatenated head outputs can be projected to a preferred dimension (hyperparameter: d_{out}).
- No worries on how #head changes dimension of outputs!



Outline of Transformer

- Add: Residual Connection
- Norm: Layer Normalization

Vaswani, Ashish, et al. "Attention is all you need." NeurIPS (2017).



Add & Norm

- Research shows that Residual Connection (He, Kaiming, et al., 2016) stabilizes training.
- Let the layers in between learn the residual relationship (i.e. $Y - X$).

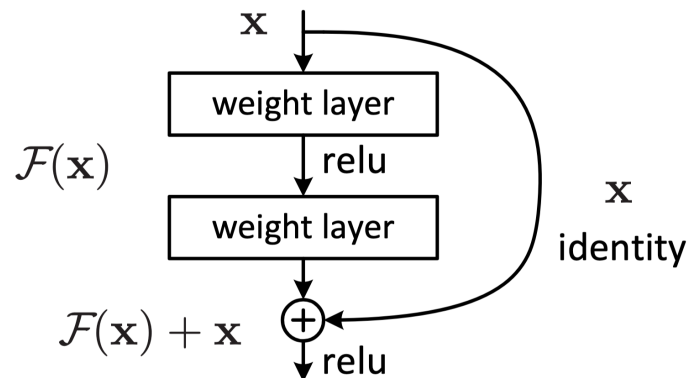
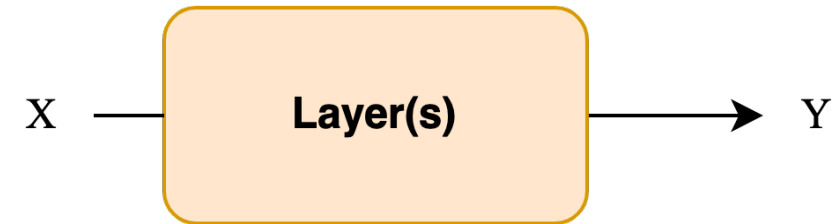
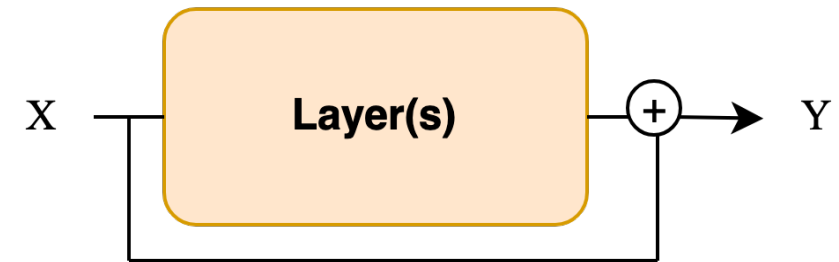


Figure 2. Residual learning: a building block.

Standard



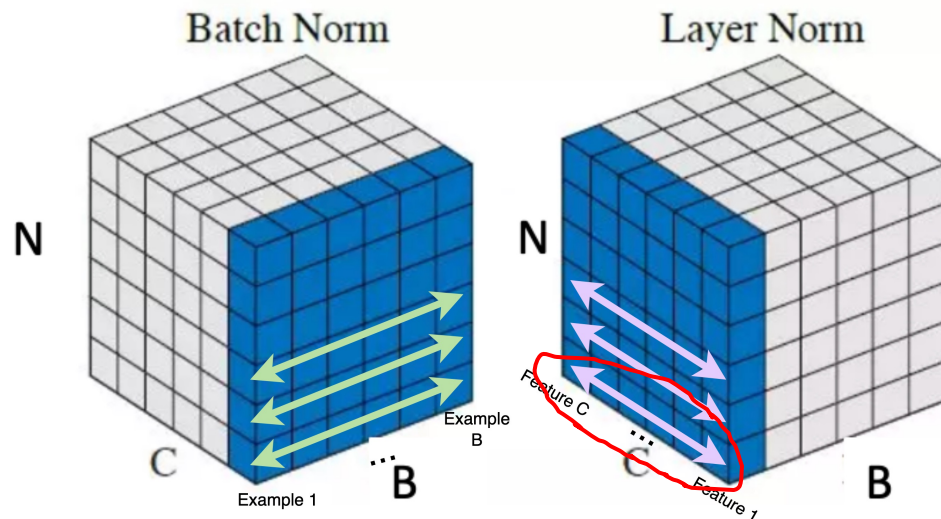
Residual



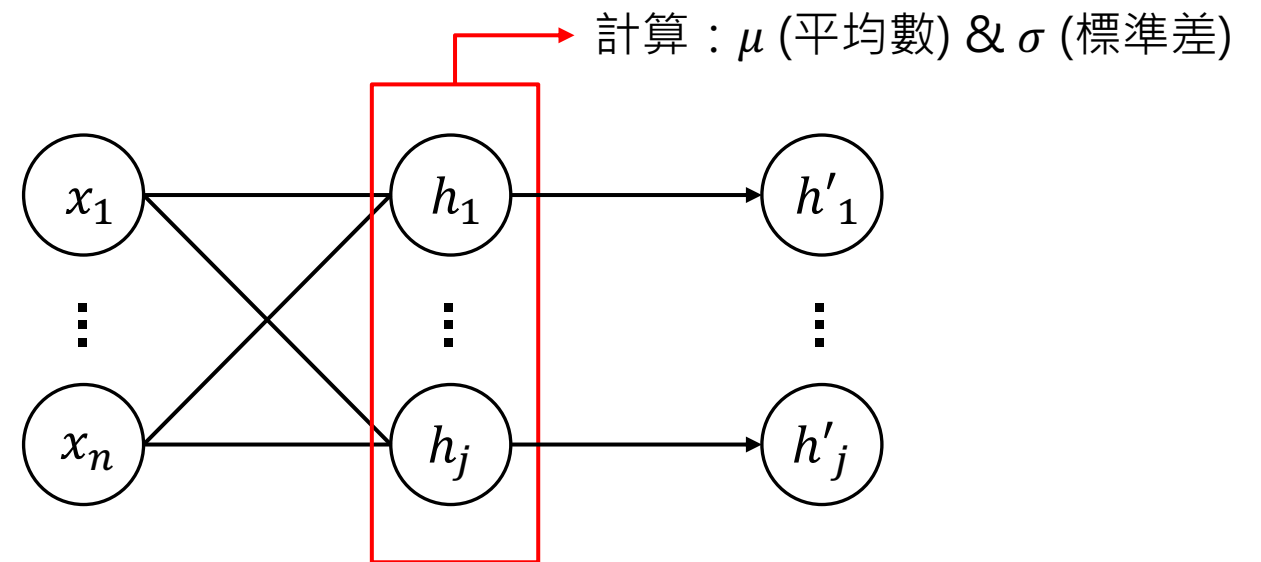
Add & Norm

- Layer Norm: For an input vector, calculate its mean and std across embedding dimension (within an example).

- $\text{norm}(X) = \frac{X - \mu}{\sigma}$

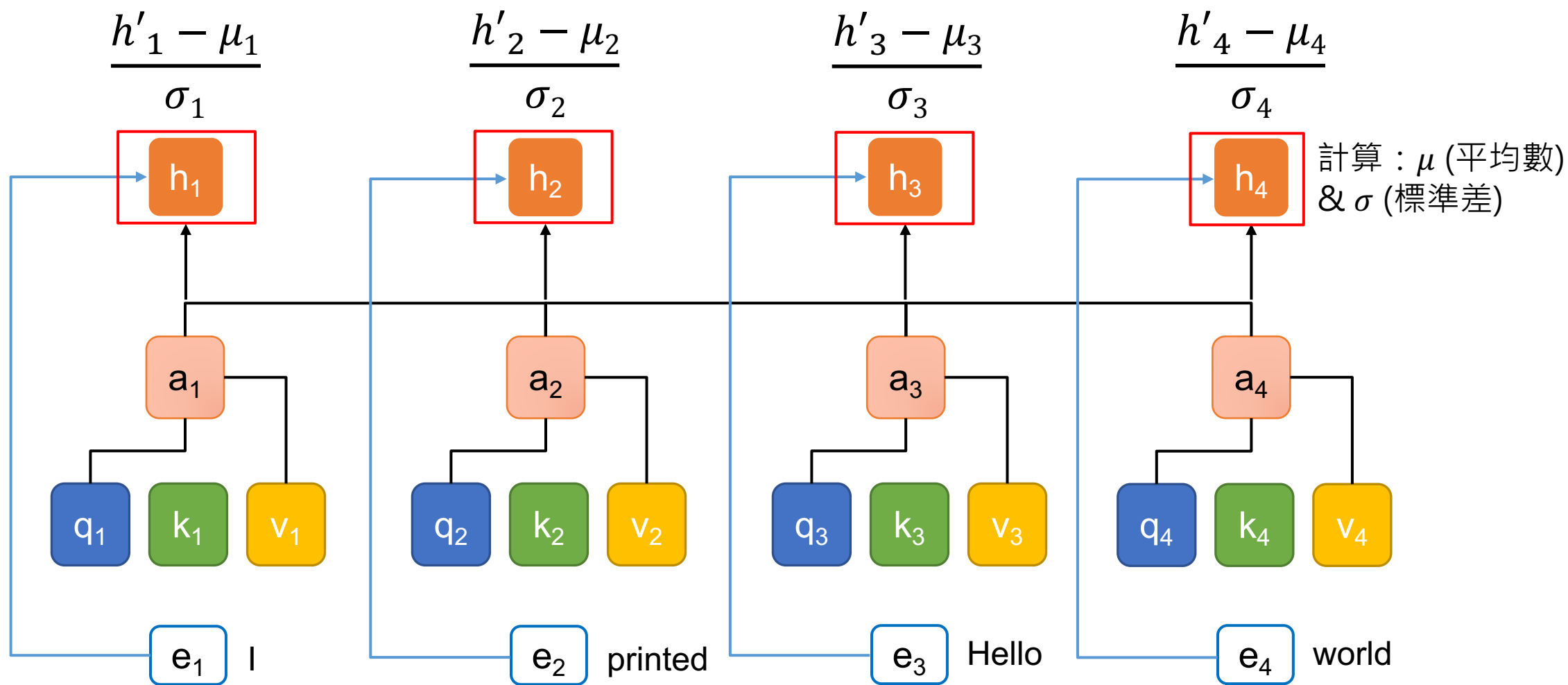


B: Batch Size
C: Channels / Embedding Feature Dimension
N: Sequence length



(Batch Normalization)

Add & Norm



Why LN not BN?

- 資料長度差距
- 模型實作表現差異

[Intro] Static padding

batch 1

I	love	CGU	[PAD]	[PAD]	[PAD]
Yesterday	I	watched	a	good	movie
Your	new	shoes	look	nice	[PAD]

longest

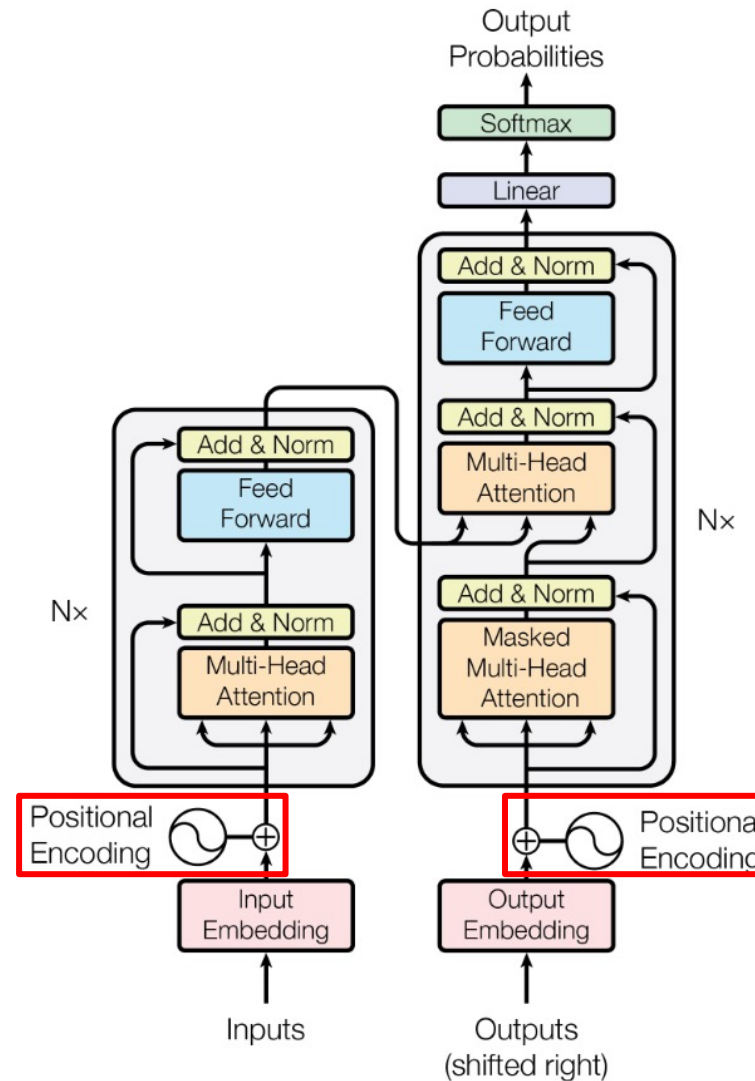
batch 2

Merry	Christmas	[PAD]	[PAD]	[PAD]	[PAD]
I	love	coding	[PAD]	[PAD]	[PAD]
Me	[PAD]	[PAD]	[PAD]	[PAD]	[PAD]



Outline of Transformer

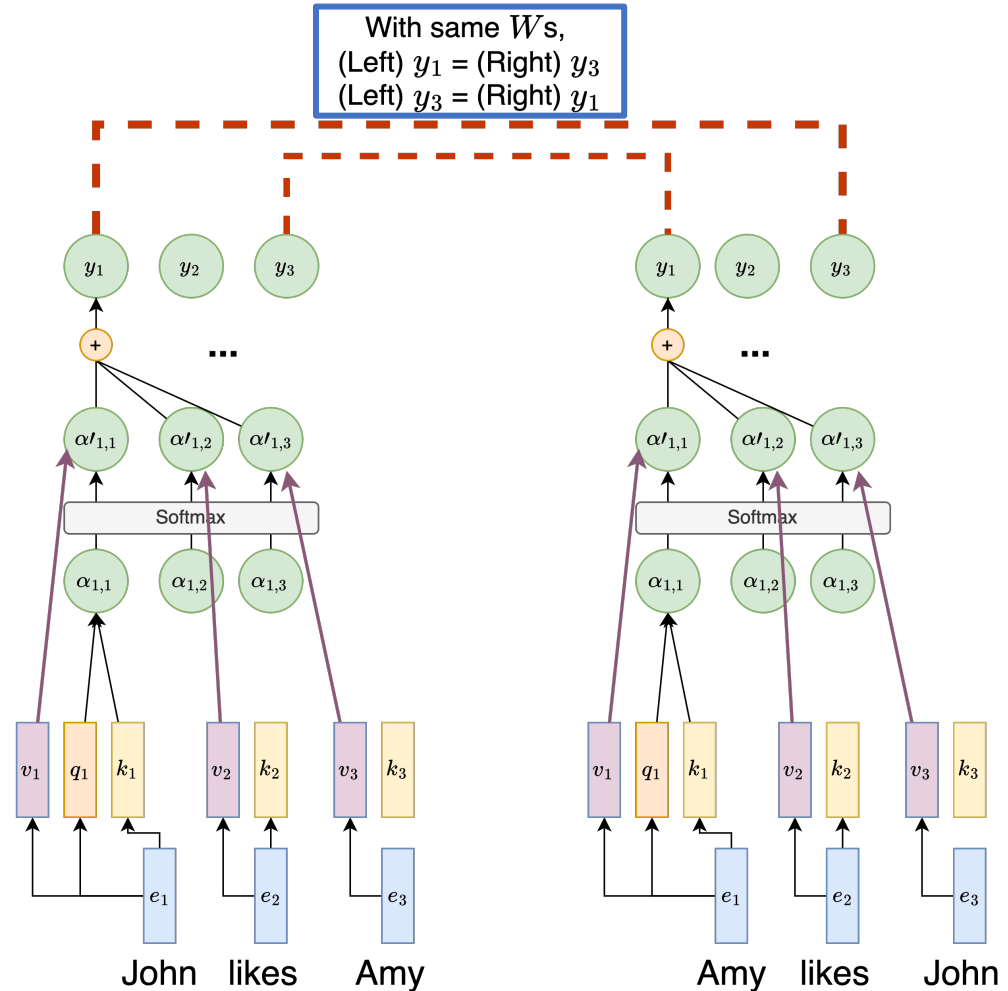
Vaswani, Ashish, et al. "Attention is all you need." NeurIPS (2017).



Lack of Position Modeling

- RNN processes sequences step-by-step.
- However, in Transformers, we have no way to know the relative distance of one word to another!
- The same word, eg. “Amy”, in the 2 sentences are encoded to the same y (with the same weight matrices).

Problematic because these 2 sentences convey different meanings, but same words are encoded identically!



Positional Encoding Equations

- 創造出相對位置的 embeddings
- 核心精神：利用三角函數的週期性來建構序列資訊
 - 偶數的 embedding 單元 ($2i$, $i = 0, 1, \dots, d_{\text{model}}$) : Sin 函數
 - 奇數的 embedding 單元 ($2i+1$, $i = 0, 1, \dots, d_{\text{model}}$) : Cosine 函數

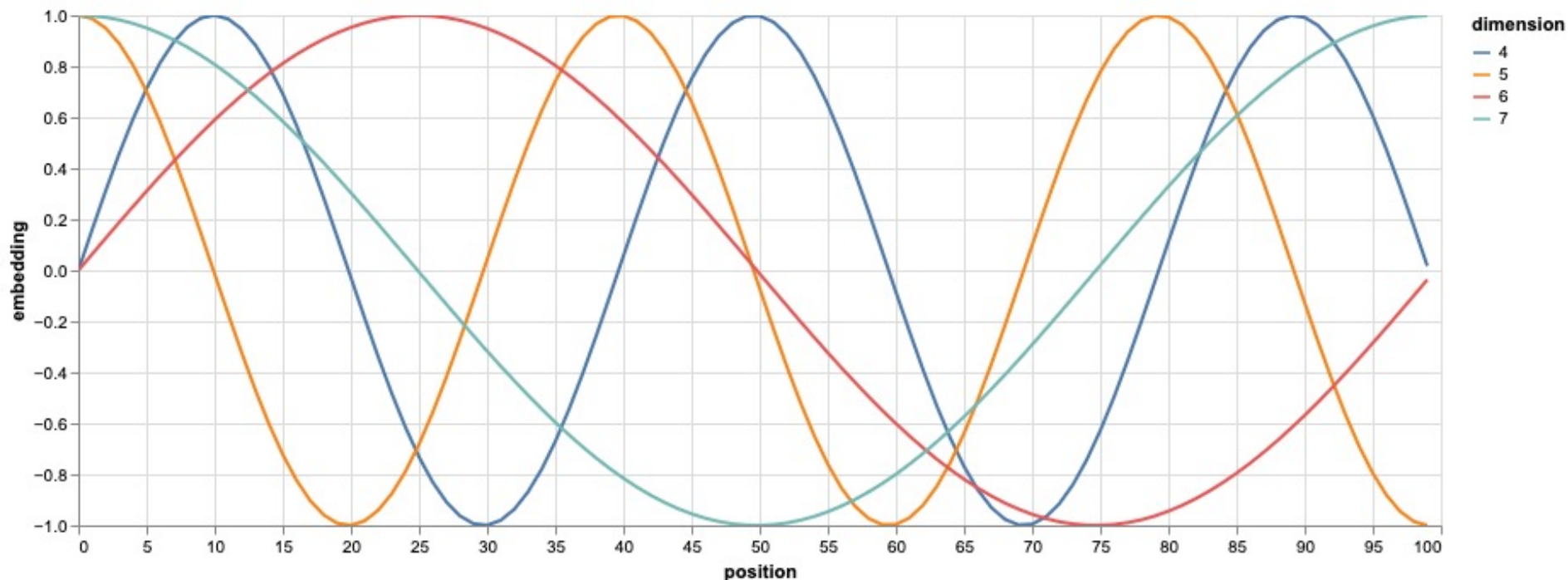
$$PE_{(pos, 2i)} = \sin(\underline{pos}/10000^{2i/d_{\text{model}}})$$

$$PE_{(pos, 2i+1)} = \cos(\underline{pos}/10000^{2i/d_{\text{model}}})$$

輸入序列中任一token的位置



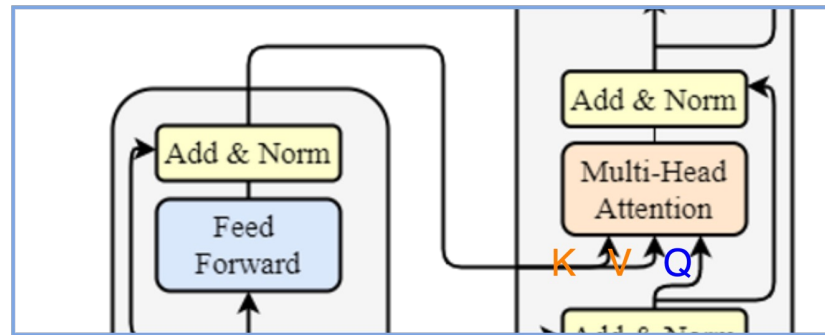
Positional Encoding Visualization



- Question: 為什麼需要同時有 Sin 和 Cosine 的函數？
- Ans: 藉由不同週期性來捕捉序列中各個位置的關係

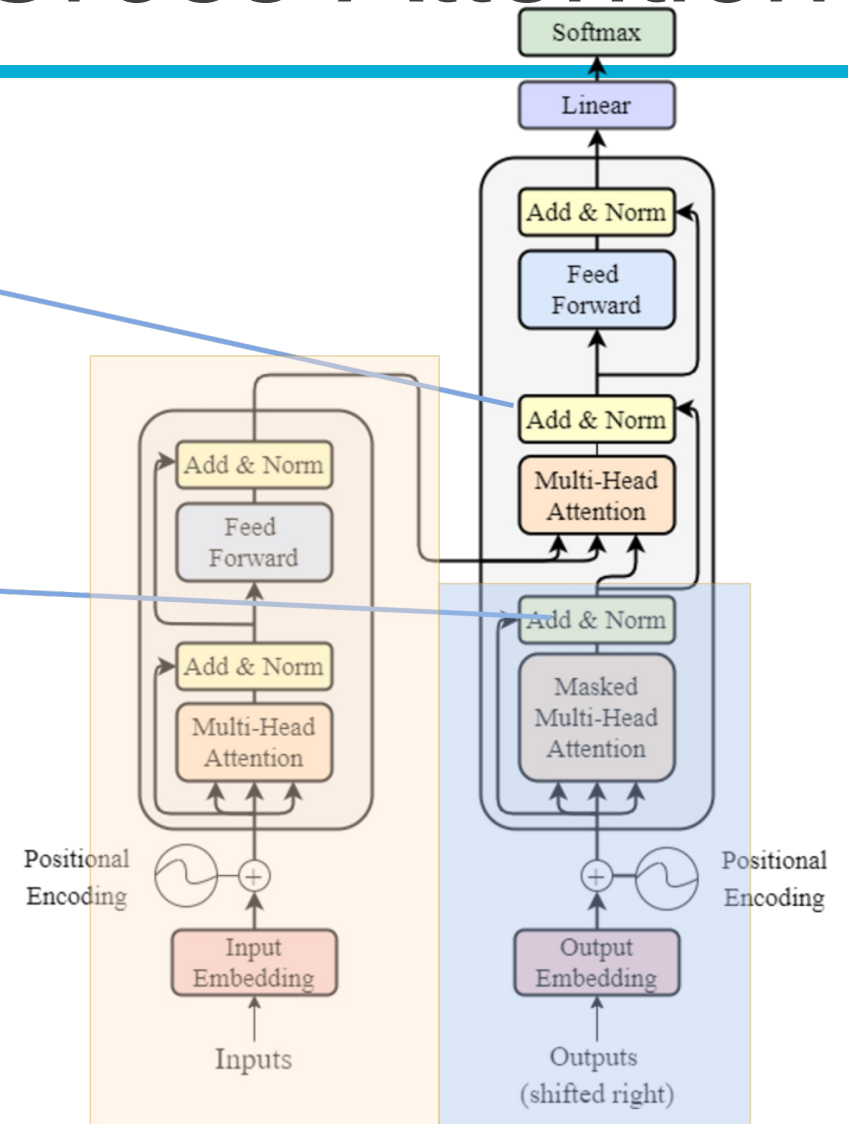


Encoder-Decoder 橋樑: Cross-Attention



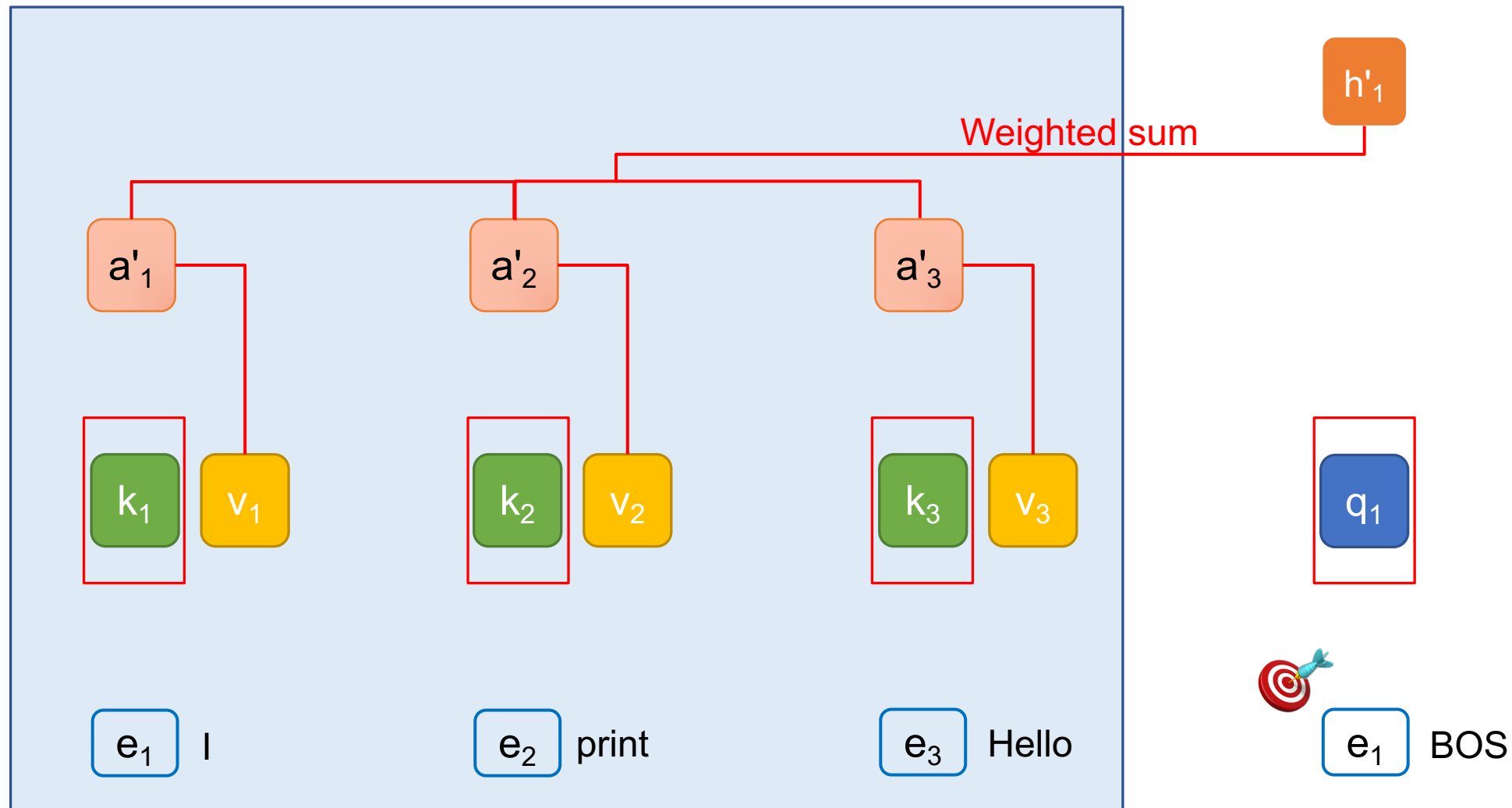
K, V uses the output from encoder as their X;
Q uses the decoder's information as X.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$



Cross-attention

Encoder



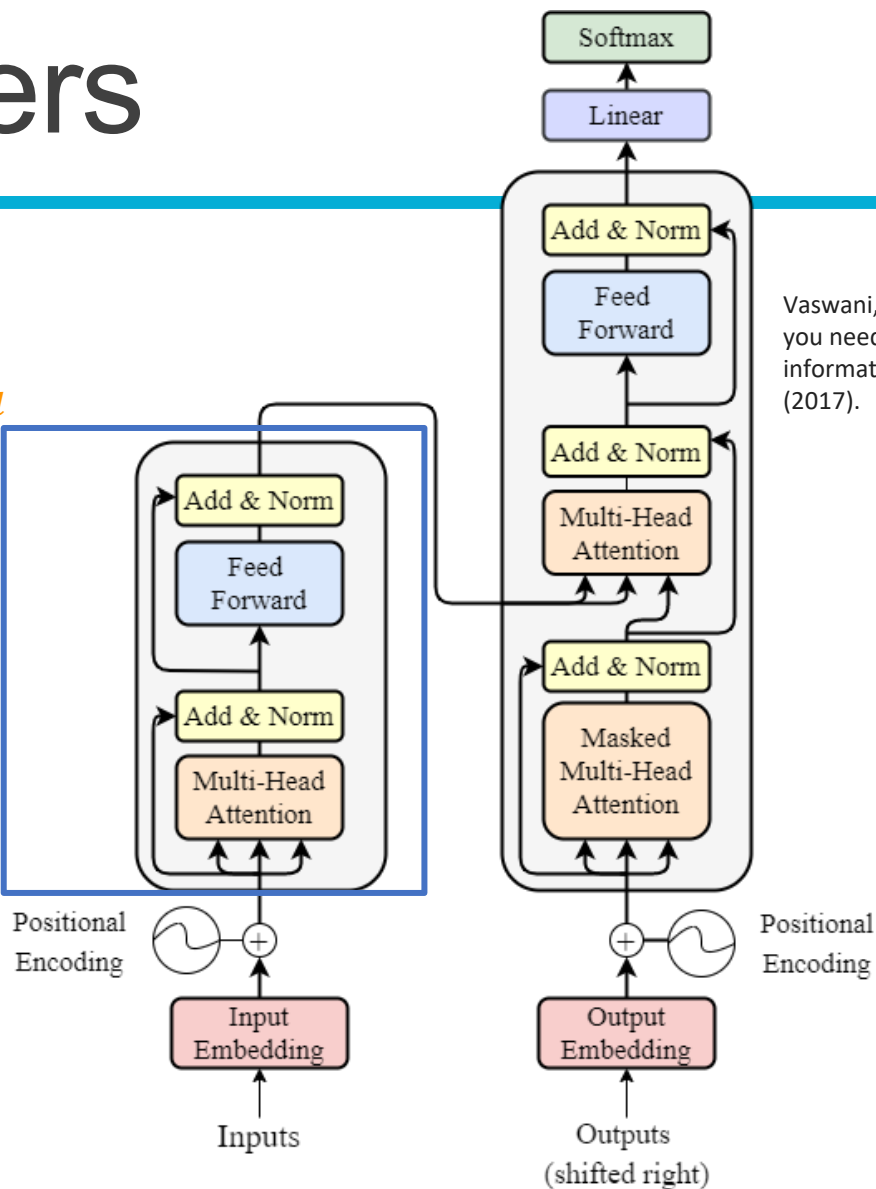
Number of Layers

Input, output dimension match!
We can have multiple encoders stacked up, to increase the number of trainable parameters.

$$\text{output} \in \mathbb{R}^{N \times d}$$

N *

$$\text{input} \in \mathbb{R}^{N \times d}$$



Vaswani, Ashish, et al. "Attention is all you need." Advances in neural information processing systems 30 (2017).

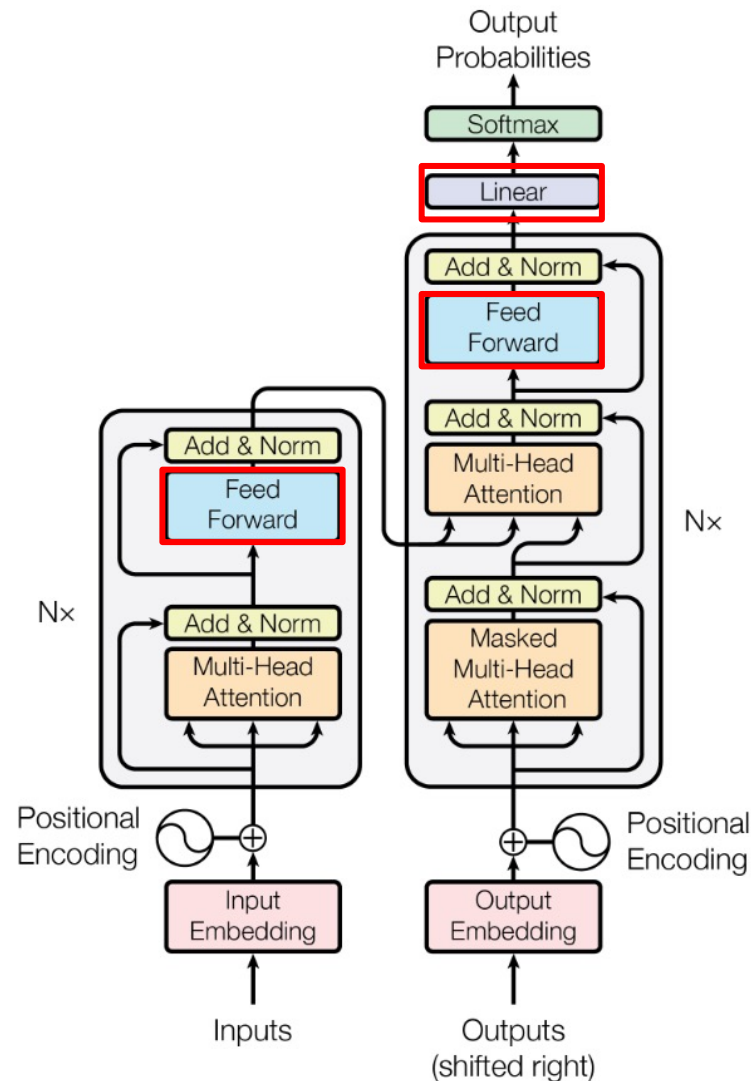


Outline of Transformer

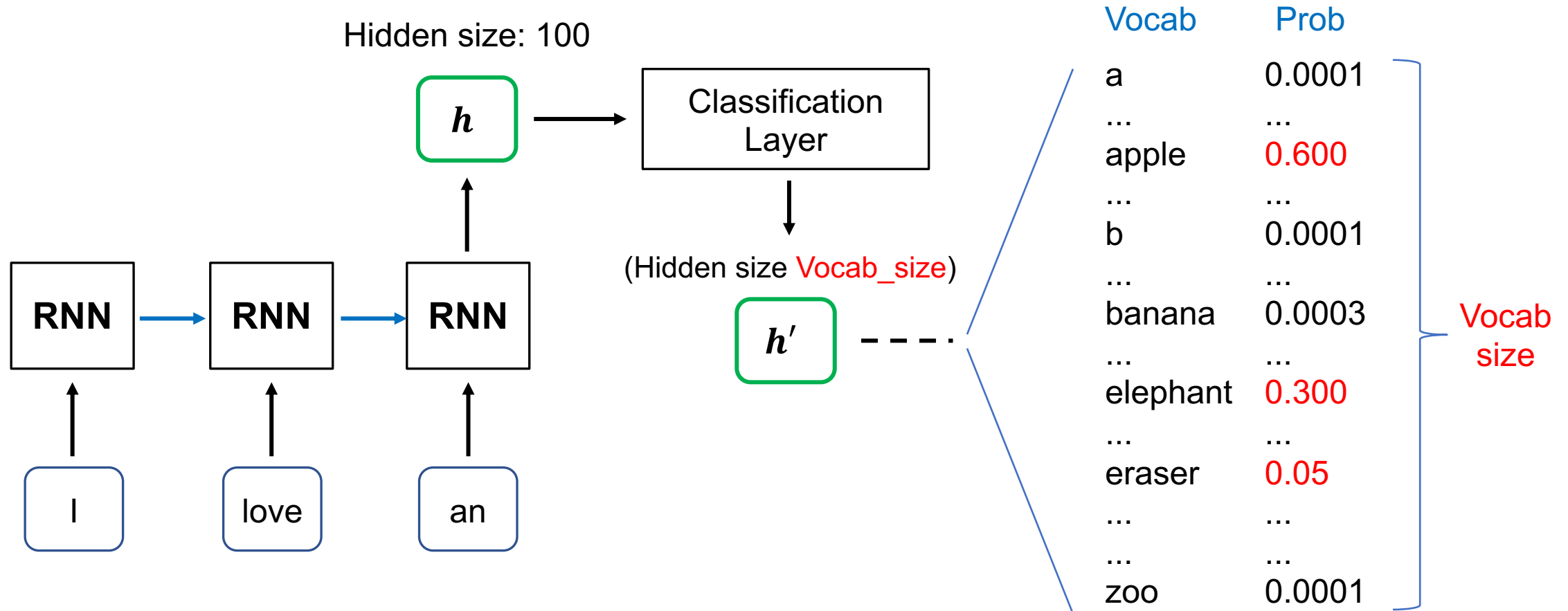
Feed Forward Networks:

- Simply multiply by a weight matrix (MLP).
- Used to project to a specific dimension.

Vaswani, Ashish, et al. "Attention is all you need." NeurIPS (2017).



[Recap] Text Generation with RNN



Transformers' Achievements

1. In original paper, State-of-The-Art (SoTA) of machine translation.

Table 2: The Transformer achieves better BLEU scores than previous state-of-the-art models on the English-to-German and English-to-French newstest2014 tests at a fraction of the training cost.

Model	BLEU		Training Cost (FLOPs)	
	EN-DE	EN-FR	EN-DE	EN-FR
ByteNet [18]	23.75			
Deep-Att + PosUnk [39]		39.2		$1.0 \cdot 10^{20}$
GNMT + RL [38]	24.6	39.92	$2.3 \cdot 10^{19}$	$1.4 \cdot 10^{20}$
ConvS2S [9]	25.16	40.46	$9.6 \cdot 10^{18}$	$1.5 \cdot 10^{20}$
MoE [32]	26.03	40.56	$2.0 \cdot 10^{19}$	$1.2 \cdot 10^{20}$
Deep-Att + PosUnk Ensemble [39]		40.4		$8.0 \cdot 10^{20}$
GNMT + RL Ensemble [38]	26.30	41.16	$1.8 \cdot 10^{20}$	$1.1 \cdot 10^{21}$
ConvS2S Ensemble [9]	26.36	41.29	$7.7 \cdot 10^{19}$	$1.2 \cdot 10^{21}$
Transformer (base model)	27.3	38.1	$3.3 \cdot 10^{18}$	
Transformer (big)	28.4	41.8	$2.3 \cdot 10^{19}$	

6 encoders,
6 decoders,
h = 512, 100K train
steps.

6 encoders,
6 decoders,
h = 1024, 300K
train steps.

Vaswani, Ashish, et al. "Attention is all you need." Advances in neural information processing systems 30 (2017).
<https://arxiv.org/abs/1706.03762>



Transformer Variants (there are many more)

- Universal Transformer

- Adds Adaptive Computation Time

- Dehghani, Mostafa, et al. "Universal transformers." arXiv preprint arXiv:1807.03819 (2018).

- Longformer

- Tackles long documents

- Iz Beltagy, et al. (2020). Longformer: The Long-Document Transformer. arXiv:2004.05150.

- Roformer

- Changes the design of positional encoding

- Su, Jianlin, et al. "RoFormer: Enhanced Transformer with Rotary Position Embedding. CoRR abs/2104.09864 (2021)." arXiv preprint arXiv:2104.09864 (2021).

- Vision Transformer

- Tackles vision tasks

- Alexey Dosovitskiy, et al. "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale." International Conference on Learning Representations. 2021.



Tokens vs. words

- token 是 (語言) 模型在每個時間點處理的單位
- word 是語言本身的單位
 - token 可以是 word，也可以是 sub-word
 - 一個 word 可以是一個 token，但單純講 token 不一定指的是 word

Traditional word tokenization

I print Hello world

Sub-word Tokenization

I prin t Hell o world

Transformer 採用此方式



Transformer 架構常用於現代 AI 模型

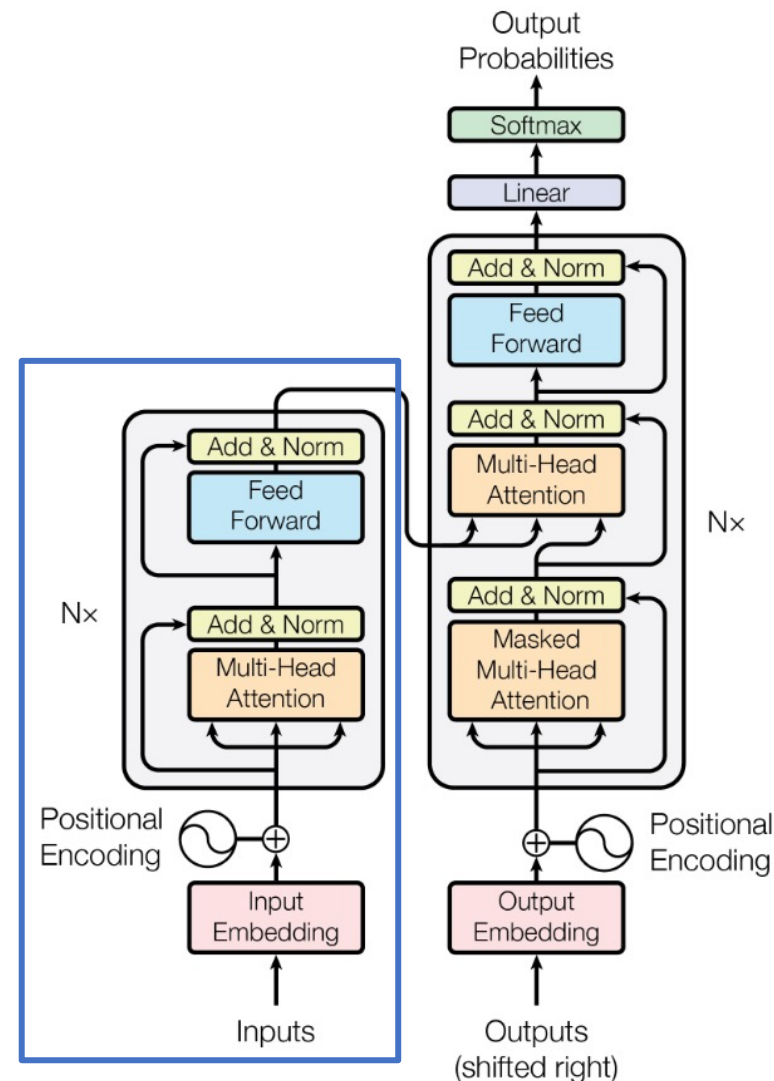
For example:

- BERT (Bidirectional encoder representations from transformers)
 - [Devlin et al., 2018](#)
- GPT series



(BERT)

N=12
Encoder



Additional Links

- <https://nlp.seas.harvard.edu/annotated-transformer/>
- <https://datascience.stackexchange.com/questions/82451/why-is-10000-used-as-the-denominator-in-positional-encodings-in-the-transformer>



Announcements

Project checkpoints

- ~~Week 9: 確定各組的題目~~
- **Week 12: 進度報告 PPT (5 pages)**
- Week 14: 進度報告 PPT (5+5 pages), Presentations (selected teams)
- Week 16: Final presentations for all teams
- Week 16-17: 繳交書面報告以及程式碼



Project 各個階段分數佔比

Final Project 佔學期總成績 40%

查核點 (週次)	對象: 繳交內容	分數佔比
Checkpoint1 (Week 12)	All teams: 進度報告 PPT (5 pages)檔案	5%
Checkpoint2 (Week 14)	All teams: 進度報告 PPT (5+5 pages*)檔案 Selected teams: 取6組 (1題目2組) 於課堂中報告, 1組 10min	5%
Checkpoint3 (Week 16)	All teams: 最終口頭報告	15%
Checkpoint4 (Week 16-17)	All teams: 書面報告檔案 (含程式碼)、小平台 (optional)	15%

*繼承Checkpoint1內容+實作



Thank you!

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